ECN 211: Intermediate Math Macro

* Lecture 1. Time Series

- investment is the most votaile in GDP
  - Consumption smoothing

Feb 15th, 2025

① Measuring the Economy: GDP

* Measuring the Economy

Def: Economic growth — a sustained increase in a nation's standard of Living, commonly measured as real GDP per capita

- Problems with real GDP per capita?

  pollution, congestion, inequality

Def: nominal GDP: total value of final businesses & services produced in an economy during a particular time period

Note: ① prices ↑ overtime so does nominal GDP

② Final — inputs don't count!

Def: Real GDP — nominal GDP measured at a given price level

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{Price level}}$$

- we are interested in the growth rate

$$g_t = \frac{Y_t - Y_{t-1}}{Y_{t-1}}$$

where g_t is growth rate from period $t-1$ to t and Y_t is some measured variable (eg. GDP) in period t

* Chapter 2

Method 1 - Expenditure Approach - sum of all expenditures spent on final goods and services

$$Y = C + I + G + X - M$$

Method 2 - income approach - sum of all income earned by factors of production

ex. 2 factors of production — labor & capital

Y = total wages (labor income) + total rent (capital income)

Method 3 - product approach - sum of value added over all firms in the economy

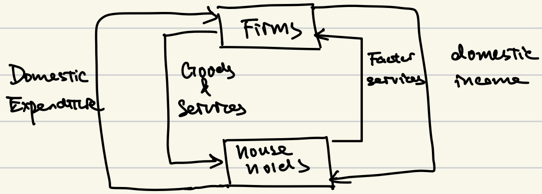
Final Goods — those sold directly to end users ex: groceries

intermediate goods — those produced by one firm, for use as an input by another firm ex. car parts

value added: difference in value of final goods and intermediates used to produce that good

* NIPA Models (National Income & Product Accounting) Measuring the "Economy"

NIPA Model 1 - closed, no government, no capital market / Household owns capital



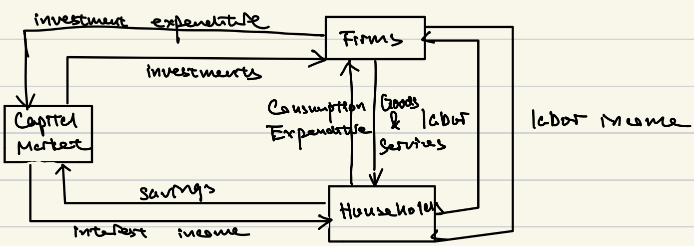
GDP = domestic income (income approach)
or
= domestic expenditure (expenditure approach) C+I

NIPA Model 2 - closed, no government, with capital market

Differences with NIPA Model 1: ① firms purchase investment goods from capital market

② Households save in capital market and gain interest

investments & savings difference



GDP = consumption expenditure + investment expenditure

(expenditure Approach) C+I

or

interest income + labor income

(income approach)

NIPA Model 4 - open, gov't, no capital market / household owns the capital

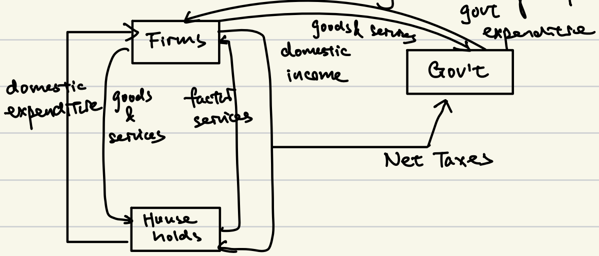
Jan 20th

NIPA Model 3 - closed, added government, no capital market / household owns capital

Differences with NIPA Model 1:

① Govt - taxes households, pays transfers to households
purchases final Goods & Services from the firms
ex. social security, military equipment

} so receives net taxes where net taxes = taxes - transfers

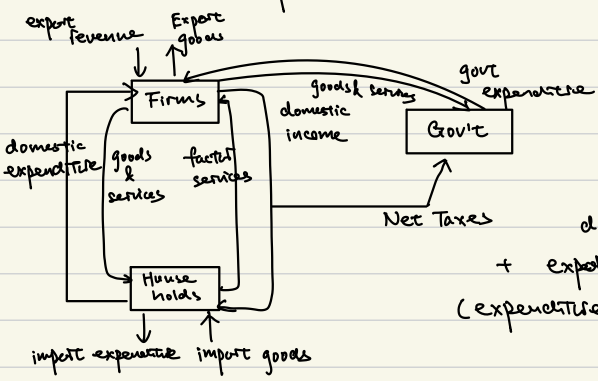


GDP = domestic expenditure + government expenditure (expenditure approach) C+I+G
or
domestic income (income approach)

NIPA Model 4 - open, gov't, no capital market / household owns the capital

Differences with NIPA Model 3:

① Household purchases imports from the rest of the world
② Firms sell exports to the rest of the world



GDP: domestic income (income approach)
or
domestic expenditure + government expenditure + export revenue - import expenditure
(expenditure approach) C + I + G + X - M

* Measuring Capital Stock

Law of motion of Capital : $K_{t+1} = (1-\delta) K_t + I_t$

depreciation   new investment

where K_t = capital stock at time t

δ = depreciation rate , $0 \le \delta \le 1$

I_t = investment at time t

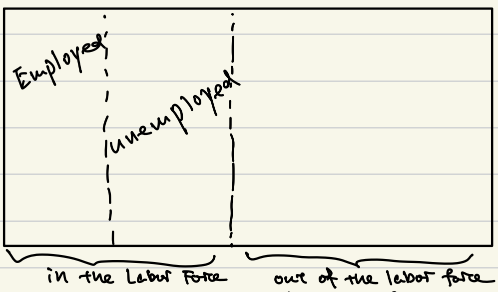
Def: Gross investment - amount of new capital created at a given time, I_t

- net investment - amount of capital added to the existing capital stock $= I_t - \delta K_t$
- net domestic product (NDP) - value of all final goods & services produced in an economy in a given period , less depreciation $\rightarrow C + I - \delta K + G + X - M$

* Measuring U.S. Unemployment

Def: Civilian Adult Population - those over 16, not institutionalized (X prison/in hospital...), not in the armed forces

CAP :



- labor force: those working or looking for work, notice this can change significantly (women)

$$\text{Labor Force Participation Rate} = \frac{\#\text{ ppl in the labor force}}{\#\text{ ppl in the C.A.P.}}$$

- Employed : those working for pay at least 1 hour in the past week (or 15 or more hours unpaid in family business) or on sick leave, or vacation

$$\text{employment rate} = \frac{\#\text{ ppl employed}}{\#\text{ ppl in the C.A.P.}}$$

- Unemployed: those not working in the past week but who looked for work in the last 4 weeks

unemployment rate $= \frac{\#\text{ ppl unemployed}}{\#\text{ ppl in labor force}}$

* Measuring Inflation

Def: Inflation is the change (increase) in the economy's overall price level

$i_t = \frac{P_t - P_{t-1}}{P_{t-1}}$ where P_t is an aggregate price level, which is commonly measured in the form of a price index.

Jan 22nd

- Common Price indices:
  - Consumer price index (CPI): weighted average price of a representative basket of consumer goods
  - Producer Price Index (PPI): weighted average price of a representative basket of producer goods

Note: PPI "leads" the CPI, ie. price changes occur in the PPI before they occur in the CPI since PPI measures input prices of consumer goods in CPI

- GDP Deflator, GDP Price Index
  - Why do we care about inflation?
    - index for contracts
    - erodes standard of living
    - distributional effects
    - distorts decisions by consumers and firms (buying gold → safety)

Chapter 4: The Classical Model

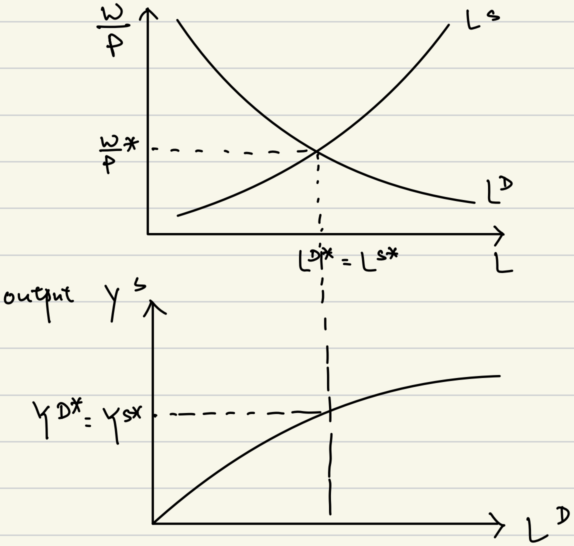
* Classical Model Roadmap

ch4
- Moving from measurement to explanation:
  - output supplied, Y^S, and demanded, Y^D
  - Labor Supplied, L^S, and labor demanded, L^D
  - Capital Market:
    - Price level, P, Wage Rate, w, rental rate, r
    - Government taxes, T
- What we want to explain / Study:
  - How much output does the economy produce (aggregate supply)?
  - Given labor is our only input,
    - how much labor do Households provide (labor supply curve)?
    - Households preferences, utility function
    - How much labor do firms hire (labor demand curve)?
      - production capacity, production function
- Using an equilibrium method, we find output, employment, and wages

ch5
- How much output does the economy demand (aggregate demand)?
- relationship between price & quantity demanded by the whole economy
- Money in the economy
- determination of the price level

ch6
- How are resources moved from savers to investors?
  - Capital Market
  - How much do HHs save?
  - How much capital do firms use?

* Chapter 4: Classical Model

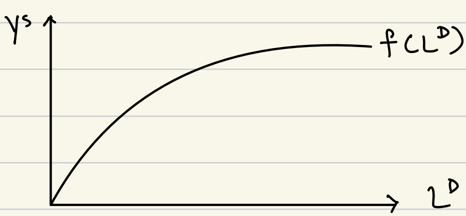


$\Rightarrow Y^{S*} = Y^{D*}$

Firms production function

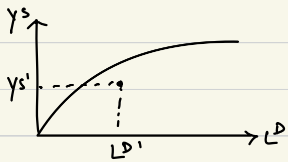
* Representative Firm

- produces good Y^S at price P
- demand labor input L^D at price W (firm's level of capital
- firm owns a technology, ie. prod func, $Y^S \leq f(L^D)$ assumed fixed)



$f'(L^D) > 0$ ie. strictly increasing
$f''(L^D) < 0$. ie. concave

- the curve $f(L^D)$ is the production frontier $(Y^S = f(L^D))$
- area on & under the curve is the production possibilities $Y^S \leq f(L^D)$



- diminishing MPL, $f''(L^D) < 0$

- assume firm produces on the frontier
- Marginal Product of Labor (MPL) $\equiv f'(L^D)$

Why? $f''(L^D) < 0$

- fixed level of Capital

* Firm max profit subject to their technological constraint

$$\Rightarrow \max_{Y^S, L^D} \; p \cdot Y^S - w \cdot L^D \quad \text{s.t. } Y^S = f(L^D)$$

$$\Rightarrow \max_{L^D} \; \underbrace{p \cdot f(L^D) - w \cdot L^D}_{L}$$

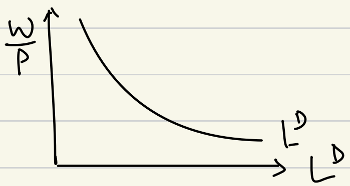
- Setup Lagrangian, solve by differentiating with respect to L^D, then set equal to zero

$$\Rightarrow L = p \cdot f(L^D) - w \cdot L^D$$

$$\Rightarrow \frac{dL}{dL^D} = p \cdot f'(L^D) - w = 0$$

$$f'(L^D) = \frac{w}{p}$$ ie. firm max profit when MPL = Real wage $\left(\frac{w}{p}\right)$

- Solve this for L^D in terms of w/p to get Labor Demand curve



* representative Household
  - consumes good Y^D at price P
  - supplies labor L^S and get wage w
  - Households have preferences in the form of a utility func, $U(Y^D, L^S)$

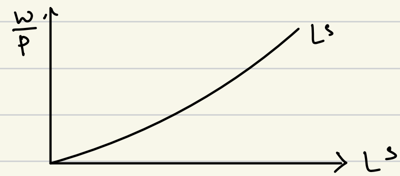
$$\frac{du(Y^D, L^S)}{dY^D} > 0 \qquad \frac{du(Y^D, L^S)}{dL^S} < 0$$ } trade off between wanting to consume and not wanting to work

  - Assume $U(Y^D, L^S)$ is concave

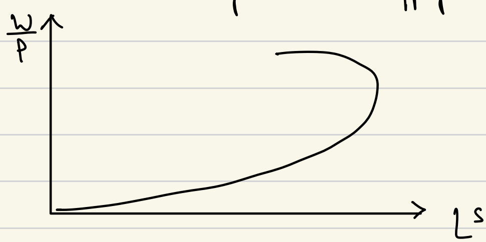
- HH max utility subject to BC

$$\Rightarrow \max_{Y^D, L^S} U(Y^D, L^S) \quad \text{s.t.} \quad p \cdot Y^D = w \cdot L^S$$

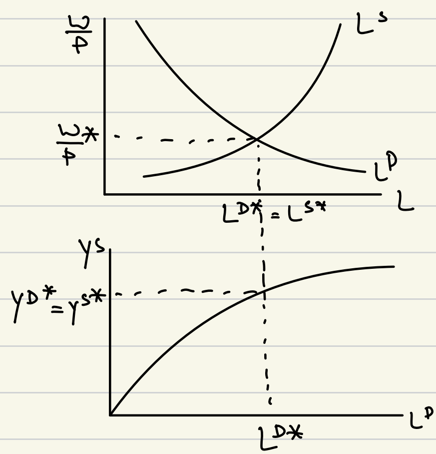
- set up Lagrangian, solve through the labor market by differentiating wrt L^S, then set equal to zero, solve L^S in terms of $\frac{w}{p}$ to get Labor supply Curve:



- Backward bending labor supply curve



* Equilibrium



- we want $\frac{w}{p}^*$ s.t. $L^D(\frac{w}{p}) = L^S(\frac{w}{p})$

- Note: we now have determined output, employment (hours worked), and wages

- Formally...
  - A competitive equilibrium in the classical model is prices (p^*, w^*) and quantities $(L^{D*}, L^{S*}, Y^{D*}, Y^{S*})$ such that the following conditions hold:
    1) Given prices (p^*, w^*) the representative HH solves:

$$\max_{Y^D, L^S} U(Y^D, L^S) \quad \text{s.t.} \quad p Y^D = w L^S$$

2) Given Prices (p^*, w^*), the representative firm solves:

$$\max_{Y^S, L^D} \; pY^S - wL^D \quad \text{s.t.} \quad Y^S = f(L^D)$$

3) Markets Clear : $Y^{S*} = Y^{D*}$ and $L^{S*} = L^{D*}$

* Chapter 5: Quantity Theory of Money (AD/AS)

- medium of exchange
- store of value
- unit of account

· Money will be cash for us

· HH demands money, M^D because of its functions

· Monetary authority creates money supply, M^S

- Assume $M^S = M^D$ ie. all money is held by someone

purchases

- M^D depends on P, Y^D, and propensity to held money, k
- Def: the propensity to hold money is the fraction of the value of G&S demanded that a HH chooses to keep in cash

$$k = \frac{M^D}{P \cdot Y^D}$$

- Note: k assumed fixed in the classical model
- Using

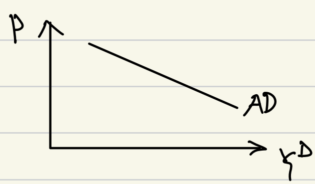
$$k = \frac{M^D}{PY^D}$$

$$P = \frac{M^D}{kY^D}$$

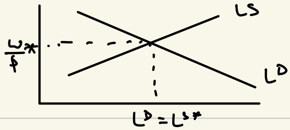
$$P = \frac{M^S}{kY^D}$$ using $M^D = M^S$ to derive classical aggregate demand curve for a given M^S and k

- Note: downward sloping as expected

$$P = \frac{M^S}{k}(Y^D)^{-1} \Rightarrow \frac{dP}{dY^D} = -\frac{M^S}{k}(Y^D)^{-2} < 0$$

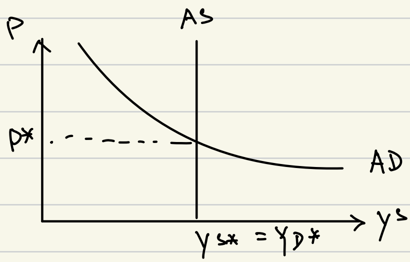


AS:



Assumption: L^*, and thus Y^*, is independent of P since equilibrium is determined by $(\frac{w}{P})^*$, assuming w non-sticky ie. can change with changes in P

⇒



Note: perfectly inelastic

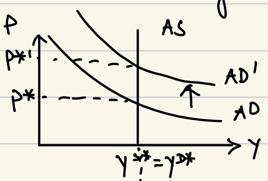
$$P = \frac{M^S}{KY^D}$$

- Def: the "neutrality of money" → nominal variables will move in proportion to changes in the quantity of money (M^S) and real variables will be unaffected

Nominal variable: Price, Wage, PY^S

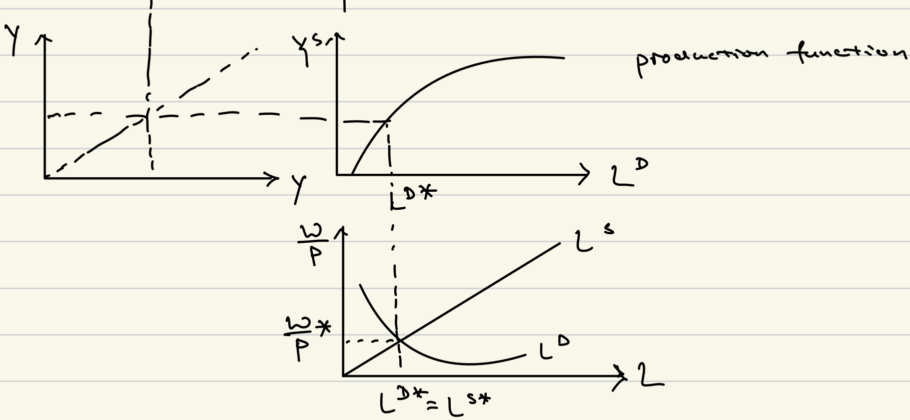
Real Variable: L^S, L^D, Y^S, Y^D, $\frac{w}{P}$

- Does neutrality of money hold in the classical model? Yes. Why?



M^S ↑ from $P = \frac{M^S}{KY^D}$

⇒ P ↑



- Note: real variables Y^*, L^*, $(W/P)^*$ do not change
  nominal variables P^*, P^*Y^*, W^* do change

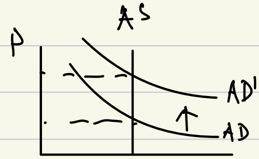
* Classical Model and Inflation

- Recall: $i_t = \dfrac{P_t - P_{t-1}}{P_{t-1}} \quad \forall t$ ie. inflation

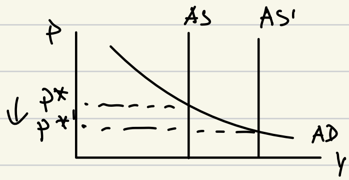
Using $P_t = \dfrac{1}{K}\dfrac{M^s_t}{Y^D_t}$ and $Y^D_t = Y^s_t$ & skipping derivation

$$i_t \approx \frac{M^s_t - M^s_{t-1}}{M^s_{t-1}} - \frac{Y^s_t - Y^s_{t-1}}{Y^s_{t-1}}$$

$=$ growth rate of M^s_t $-$ growth rate of Y^s_t



$\dfrac{di_t}{dg^{M^s}_t} > 0$ if $g^{M^s}_t > 0$ ie. inflation $\uparrow$ if $M^s \uparrow$



$\dfrac{di_t}{dg^{Y^s}_t} < 0$ if $g^{Y^s}_t > 0$ ie $i_t \downarrow$ if $Y^s_t \uparrow$

Feb 5 th.

* chapter 6: savings & investment

* Dynamic, Two period Classical Model with Capital Market
  - two time periods, $t = 1, 2$
  - HH and firm totally separate agents (do not come together in the goods mkt but come together in the capital mkt)
  - present value: \$1 today is more valuable than \$1 tomorrow ... why?
    $0 \leq r \leq 1 \rightarrow$ \$1 today earns interest r
  - discount factor β, $0 \leq \beta \leq 1 \rightarrow$ weight given future consumption
    $\beta = 0 \Rightarrow$ "impatient", no value on the future
    $\beta = 1 \Rightarrow$ "patient", value on the future
    ex/: $u(c_1, c_2) = c_1 + \beta c_2$

* Representative HH
  - utility func $U(c_1, c_2)$
  - $c_1 \rightarrow$ consumption in period 1, $c_2 \rightarrow$ consumption in period 2
  - Labor inelastically supplied, ie: $L^S = L^D = 1$, so does not enter the utility function & is not chosen by the HH; wages are not paid for labor services
  - Lucas Tree
  - set $p^* = 1$, everything real

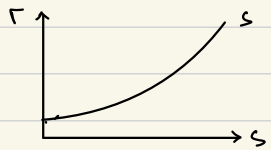
* Period 1
  - HH receives endowment of goods Y_1 from the Lucas Tree
  - chooses amount c_1 to consume & amount of savings S where $S = Y_1 - c_1$

$\Rightarrow B^c_{t=1}$: $c_1 + S = Y_1$

* Period 2
  - HH receives endowment of goods Y_2 from Lucas Tree
  - HH receives return on period 1 savings, $(1+r)S$
  - choose amt c_2 to consume but not amt of savings S (no period 3)

$$\Rightarrow B^c_{t=2} = c_2 = Y_2 + (1+r)S$$

* Timing
  - HH knows Y_1 and Y_2 in period $t=1$
  - HH chooses C_1, C_2 and S in period $t=1$
* HH maximizes utility subject to budget constraints → savings supply curve



* Representative firm
  - prod func $Y^S = f(K, L^D)$
  - inelastic labor, $L^D = 1$; wages are not paid $\Rightarrow Y^S = f(K)$

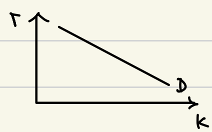
<u>Period 1</u>

- firm does not produce
- firm chooses level of capital, K

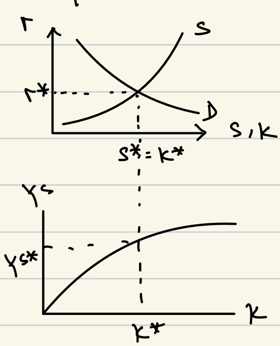
<u>Period 2</u>

- K cost $(1+r)$ per unit, paid to the HH
- firm produces $Y^S = f(K)$
- Firm maximizes profit, subject to tech constraint

  $\Rightarrow$ investment demand curve



* Equilibrium



$\Rightarrow r^*, S^*, K^*$

$\Rightarrow Y^{S*}$

Note: Y_1, Y_2 & β are not equi values they are parameters

- Uses S^* to determine C_1^* & C_2^*

- formally...

A competitive equilibrium in the dynamic, two period classical model with a capital market is a price (r^*) & quantities $(c_1^*, c_2^*, s^*, k^*, Y^{s*})$

1) Given price (r^*), the representative HH solves:

$$\max_{c_1, c_2, s} u(c_1, c_2) \quad \text{s.t.} \quad c_1 + s = Y_1$$
$$c_2 = Y_2 + (1+r^*)s$$

2) Given the price (r^*), the representative firm solves:

$$\max_{Y^s, k} Y^s - (1+r^*)k \quad \text{s.t.} \quad Y^s = f(k)$$

3) Markets Clear: $s^* = k^*$

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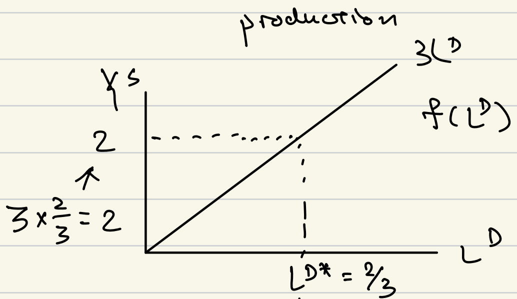


ex. $u(Y^D, L^S) = 2\ln Y^D + \ln(1-L^S)$

$Y^s = 3L^D$

Firms: Solve for $\frac{w}{p}^*$, $L^{s*}, L^{D*}, Y^{s*}, Y^{D*}$

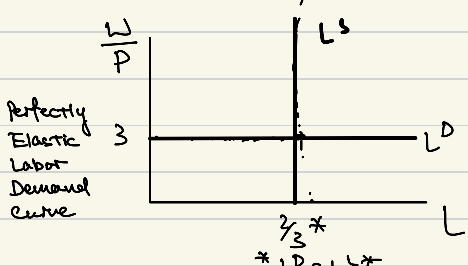
Max $p^* Y^s - w^* L^D$ given $Y^s = f(L^D)$

$\mathcal{L} = \max\ p \cdot 3L^D - w \cdot L^D$

$\delta p Y^s - w^* L^D$

$0 < \delta < 1$

$$\frac{d\mathcal{L}}{dL^D} = 3p - w$$
$$3p - w = 0$$
$$3p = w$$
$$\frac{w}{p} = 3$$



$^*L^D = L^{s*}$

Rep. Customer:

Max $u(Y^D, L^S)$ s.t. $w^* L^s = p^* Y^D$

$$\frac{w}{p} L^s = Y^D$$

Max $2\ln Y^D + \ln(1-L^s)$

$$2\ln \frac{w}{p} L^s + \ln(1-L^s)$$

$$2\frac{W}{P} \cdot \frac{1}{\frac{w}{p}L^s} - \frac{1}{1-L^s} = 0$$

$$\frac{2\cancel{W}}{\cancel{P}} \cdot \frac{\cancel{P}}{\cancel{W}L^s} = \frac{1}{1-L^s}$$

$$\frac{2}{L^s} = \frac{1}{1-L^s}$$

$$2 - 2L^s = L^s$$

$$3L^s = 2$$

$$L^s = 2/3$$

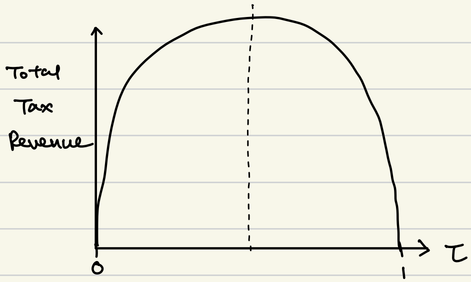
By Market Clearing, we know that $L^{D*} = L^{s*} = 2/3$

By production func, $Y^{s*} = 3L^{D*} = 3 \times \frac{2}{3} = 2$

By MKT Clearing, we know that $Y^{D*} = Y^{s*} = 2$

* Classical Model with Taxes and the Laffer Curve

Def: Laffer Curve shows the relationship between the tax rate τ and total tax revenue



Practice Problems

HH: $\max_{Y^D, L^S} U(Y^D, L^S)$ s.t. $p \cdot Y^D = w \cdot L^S$

Firm: $\max_{Y^S, L^D} p \cdot Y^S - w \cdot L^D$ s.t. $Y^S = 2kL^D$

(a) Marginal Utility of Consumption: $\frac{dU}{dY^D} = [\ln(Y^D) + \ln(1-L^S)]'$

$$= \frac{1}{Y^D}$$

(b) Marginal Utility of Labor: $\frac{dU}{dL^S} = -\frac{1}{1-L^S}$

(c) Marginal Product of Labor: $\frac{dY^S}{dL^D} = 2k$

(d) HH: $\max_{Y^D, L^S} U(Y^D, L^S)$ s.t. $p \cdot Y^D = w \cdot L^S$

Firm: $\max_{Y^S, L^D} p \cdot Y^S - w \cdot L^D$ s.t. $Y^S = 2kL^D$

(e) HH: $p \cdot Y^D = w \cdot L^S$ $\rightarrow$ $\frac{d}{dL^S}\left(\ln\left(\frac{w}{p}L^S\right) + \ln(1-L^S)\right)$

$Y^D = \frac{w}{p}L^S$

$$= \cancel{\frac{w}{p}} \cdot \frac{1}{\cancel{\frac{w}{p}}L^S} - \frac{1}{1-L^S} = 0$$

$$\frac{1}{L^S} = \frac{1}{1-L^S}$$

$$1 - L^S = L^S$$

$$L^S = \frac{1}{2}$$

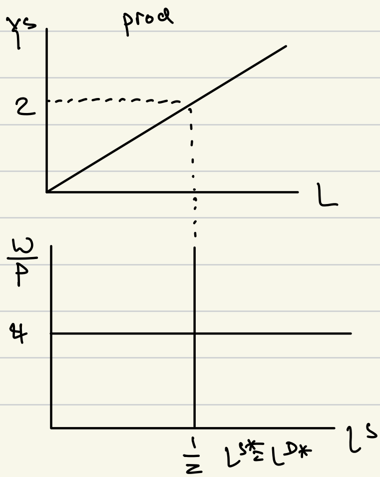
Firm: $Y^S = 4L^D$

$\frac{d}{dL^D}(p \cdot 4L^D - w \cdot L^D)$

$= 4p - w = 0$

$w = 4p$

$\frac{w}{p} = 4$



(f) $L^D = \frac{1}{2}$ $Y^S = 2$

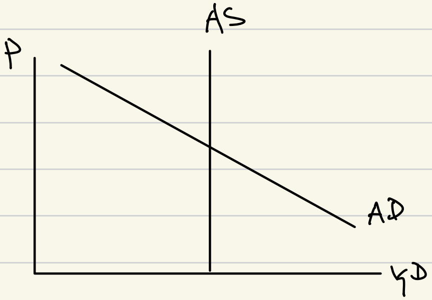
$p \cdot 4L^D - w \cdot L^D$

$= 2p - \frac{1}{2}w = 2p - \frac{1}{2} \cdot 4p = 0$

$w = 4p$

(g) $AD \Rightarrow p = \frac{M^S}{kY^D} = \frac{2}{Y^D}$

$= \frac{2}{Y^D} = \frac{2}{2} = 1$



Question 2: $U(Y^D, L^S) = Y^D - \frac{1}{2}(L^S)^2$

$Y^S = 5L^D - \frac{1}{2}(L^D)^2$

labor income $= wL^S(1-\tau)$

$5pL^D - \frac{p}{2}L^{D2}$

$MU_c = \frac{dU}{dY^D} = 1$

$\frac{d}{dL^D}\left(p(5L^D - \frac{1}{2}L^{D2}) - wL^D\right)$

$MU_L = \frac{dU}{dL^S} = -L^S$

$= 5p - pL^D - w = 0$

$MP_L = \frac{dY^S}{dL^D} = 5 - L^D$

$p(5 - L^D) = w$

$5 - L^D = \frac{w}{p}$

$L^D = -\frac{w}{p} + 5$

$wL^S(1-\tau) = p \cdot Y^D$

$wL^S(1-\tau) = p \cdot Y^D$

$Y^D = \frac{w}{p}(1-\tau)L^S$

$\frac{d}{dL^S}\left(\frac{w}{p}(1-\tau)L^S - \frac{1}{2}L^{S2}\right)$

$= \frac{w}{p}(1-\tau) - L^S$

$L^S = \frac{w}{p}(1-\tau)$

$-\frac{w}{p} + 5 = \frac{w}{p}(1-\tau)$

$5 = \frac{w}{p}(1-\tau+1)$

$5 = \frac{w}{p}(2-\tau)$ $\quad \frac{w}{p} = \frac{5}{2-\tau}$

i. Real Tax Revenue

h. $\frac{5}{2-\tau}(1-\tau) = \frac{5(1-\tau)}{2-\tau}$

i. $L^* \cdot \frac{w^*}{p^*} \cdot \tau$

$L^* = 5 \cdot \frac{1-\tau}{2-\tau} \frac{5}{2-\tau} \cdot \tau$

$= \frac{25(1-\tau)}{(2-\tau)^2}\tau$

j.

Q3. HH: $U(Y^D, L^S) = Y^D - \frac{1}{2}(L^S)^2$

Firm: $Y^S = 3L^D$

a) $pY^S \cdot \tau = T$

b) $p \cdot Y^D = L^S \cdot w + T$

$Y^D = \frac{w}{p}L^S + \frac{T}{p}$

$(\frac{w}{p}L^S + \frac{T}{p} - \frac{1}{2}L^{S2})'$

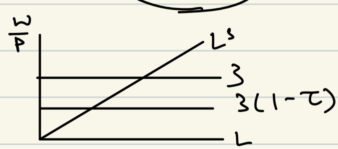
$\frac{w}{p} - L^S = 0$ $L^S = \frac{w}{p}$

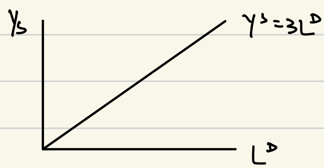
c) $(p \cdot 3L^D(1-\tau) - w \cdot L^D)'$

$= 3p(1-\tau) - w = 0$

$w = 3p(1-\tau)$

$\frac{w}{p} = 3(1-\tau)$





d) $\frac{w}{p} = L^S = L^D = 3(1-\tau)$

$Y^S = Y^D = 9(1-\tau)$

e) $Y^{D*} = Y^{S*}$

$(p \cdot 3L^D - wL^D)'$ $3(1-\tau)$

$3p = w$

$\frac{w}{p} = 3$

More

$U(Y^D, L^S) = 2\ln Y^D + 4\ln(1-L^S)$

$PY^D = wL^S$

$Y^D = \frac{w}{P}L^S \quad \rightarrow (2\ln \frac{w}{P}L^S + 4\ln(1-L^S))'$

$= 2\cancel{\frac{w}{P}} \frac{1}{\cancel{\frac{w}{P}}L^S} - \frac{4}{1-L^S}$

$\frac{2}{L^S} = \frac{4}{1-L^S}$

$2 - 2L^S = 4L^S$

$2 = 6L^S$

$L^S = \frac{1}{3}$

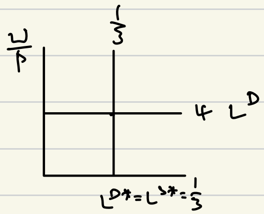
$Y^S = 4L^D$

$(P4L^D - wL^D)'$

$= 4P - w = 0$

$\frac{w}{P} = 4$

$L^D = 4$



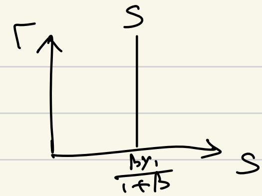
ex.

ex. $u(c_1, c_2) = \ln c_1 + \beta \ln c_2$

$Y^s = 2K^{\frac{1}{2}}$

Assume $Y_2 = 0$

HH: $S = \frac{\beta Y_1}{1+\beta}$



$Y^s = 2K^{\frac{1}{2}}$

$$[Y^s - (1+r)K]$$

$$\downarrow$$

$$\frac{d}{dK}[2K^{\frac{1}{2}} - (1+r)K]$$

$$= K^{-\frac{1}{2}} - (1+r) = 0$$

$$\frac{1}{K^{\frac{1}{2}}} = (1+r)$$

$$K^{\frac{1}{2}} = \frac{1}{(1+r)} \quad K = \frac{1}{(1+r)^2}$$

Markets Clear: Find r^*

$$\frac{\beta Y_1}{1+\beta} = \frac{1}{(1+r)^2}$$

$$(1+r)^2 = \frac{1+\beta}{\beta Y_1}$$

$$1+r = \sqrt{\frac{1+\beta}{\beta Y_1}}$$

$$r^* = \sqrt{\frac{1+\beta}{\beta Y_1}} - 1$$

$$Y^s = 2\cdot\sqrt{\frac{\beta Y_1}{1+\beta}}$$

$$\frac{Y_1 + \cancel{\beta Y_1} - \cancel{\beta Y_1}}{1+\beta} = \frac{Y_1}{1+\beta}$$

$$c_1 + S = Y_1 \qquad \|$$

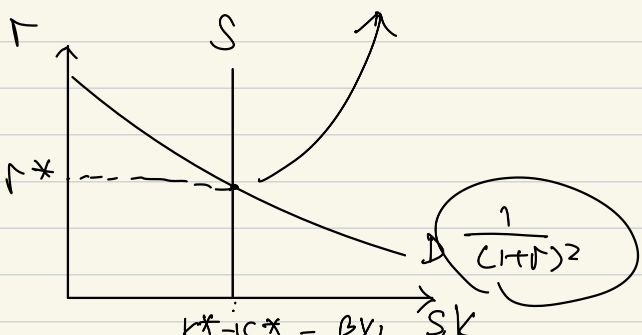
$$c_1 = Y_1 - \frac{\beta Y_1}{1+\beta}$$

$$c_2 = +(1+r^*)\cdot S$$

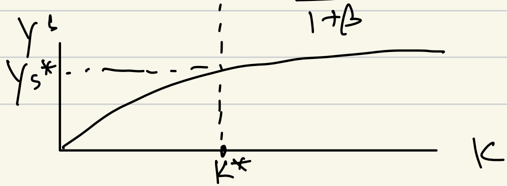
$$= \sqrt{\frac{1+\beta}{\beta Y_1}} \cdot \frac{\beta Y_1}{1+\beta}$$

$$= \sqrt{\frac{\beta Y_1}{1+\beta}}$$

$$\frac{dK}{dr} = -2(1+r)^{-3}$$



$$K^* = S^* = \frac{\beta Y_1}{1+\beta}$$



How do r^*, s^*, k^*, Y^{s*}, C_1^* & C_2^* Δ with Δ in β & Y_1?

$\beta \uparrow \Rightarrow$ HH cares more abt future

Savings Supply Curve shifts to the right

Mkts Clearing: $K = S$

Through Prod func: $Y^S \uparrow$ as well

Savings: $\frac{ds^*}{d\beta} = \frac{Y_1}{(1+\beta)^2} > 0$ ie. more HH cares about the future, the more it saves

$\frac{ds}{dY_1} = \frac{\beta}{1+\beta} > 0$ ie. "level effect" more income HH has, the more it saves

Practice Problems

$U(c_1, c_2) = c_1 + \beta c_2$

$Y_2 = 0$

a) $\text{Max } c_1 + \beta c_2 \text{ s.t. } c_1 = Y_1 - S, \; c_2 = Y_2 + (1+r)S$

$\text{Max } Y^s - (1+r)K \quad \text{s.t. } Y^s = 10K^{\frac{1}{2}}$

b. HH: $\frac{d}{dS}\left[Y_1 - S + \beta(1+r)S\right]$

$-1 + \beta(1+r) = 0$

$\beta(1+r) = 1$

$\frac{4}{5}(1+r) = 1$

$1+r = \frac{5}{4}$

$r = \frac{1}{4}$

Firm: $Y^s - (1+r)K$

$\frac{d}{dK}\left[10K^{\frac{1}{2}} - (1+r)K\right]$

$5K^{-\frac{1}{2}} = (1+r)$

$K^{\frac{1}{2}} = \frac{5}{(1+r)}$

$K = \frac{25}{(1+r)^2}$

$K = \frac{25}{(1+\frac{1}{4})^2} = 25 \cdot \frac{16}{25}$

$= 16$

$S^* = K^* = 16$

$c_1 = 25 - 16 = 9$

$c_2 = 0 + (1 + \frac{1}{4})16$

$= \frac{5}{\cancel{4}} \times \cancel{16}\,4 = 20$

$Y^s = 10\sqrt{16} = 40$

b) HH: max $\ln c_1 + \beta \ln c_2$ s.t. $c_1 = y_1 - s$, $c_2 = y_2 + (1+r)s$

Firm: Max $y^s - (1+r)k$ s.t. $y^s = 20k^{\frac{1}{2}}$

HH:

$$\ln(y_1 - s) + \beta(\ln y_2 + (1+r)s)$$

$$= \left[\ln(50-s) + \tfrac{3}{4}\ln(1+r)s\right]'$$

$$-\frac{1}{50-s} + \frac{3}{4}\frac{\cancel{(1+r)}}{\cancel{(1+r)}s} = 0$$

$$\frac{3}{4\cancel{(1+r)}s} = \frac{1}{50-s}$$

$$\frac{3}{4s} = \frac{1}{50-s}$$

$$4s = 150 - 3s$$

$$7s = 150$$

$$s = \frac{150}{7}$$

Firm:

$$\max\ [20k^{\frac{1}{2}} - (1+r)k]\frac{d}{dk}$$

$$= 10k^{-\frac{1}{2}} - (1+r) = 0$$

$$k^{-\frac{1}{2}} = \frac{(1+r)}{10}$$

$$k^{\frac{1}{2}} = \frac{10}{(1+r)}$$

$$k = \frac{100}{(1+r)^2}$$

$$\frac{100}{(1+r)^2} = \frac{150}{7}$$

$$\frac{14}{3}\quad \frac{\cancel{700}}{\cancel{150}} = (1+r)^2$$

$$r = \sqrt{\frac{14}{3}} - 1$$

$$\ln(50(1-\tau) - s) + \frac{3}{4}(\ln \quad + (1+r)s)$$

$$= \frac{-1}{(50-\tau)-s} + \frac{3}{4}\frac{\cancel{(1+r)}}{\cancel{(1+r)}s} = 0$$

$$\frac{3}{4s} = \frac{1}{50 - \cancel{50\tau} - s}$$

$$150 - 150\tau - 3s = 4s$$

$$7s = 150(1-\tau)$$

$$s = \frac{150}{7}(1-\tau)$$

$$\frac{150}{7}(1-\tau) = \frac{100}{(1+r)^2}$$

$$\sqrt{\frac{700}{150(1-\tau)}} - 1 = r$$

$$\frac{dY^{s*}}{d\beta} = \left(\frac{\beta Y_1}{1+\beta}\right)^{-1/2} \frac{Y_1}{(1+\beta)^2} > 0$$

$$\frac{dY^{s*}}{dY_1} = \left(\frac{\beta Y_1}{1+\beta}\right)^{-1/2} \frac{\beta}{1+\beta} > 0$$

more income the HH has, or more HH cares abt the future, the more HH saves; by mkt clearing, capital increases; by prod func, output supply increases

$$\frac{dr^*}{d\beta} = -\frac{1}{2}\left(\frac{\beta Y_1}{1+\beta}\right)^{-\frac{3}{2}} \left[\frac{Y_1}{(1+\beta)^2}\right] < 0$$

$$\frac{dr^*}{dY_1} = -\frac{1}{2}\left(\frac{\beta Y_1}{1+\beta}\right)^{-3/2} \frac{\beta}{1+\beta} < 0$$

ie. more income HH has or more HH cares abt the future, the more HH saves; pushing out savings supply curve, driving down the interest rate

$$\frac{dc_1^*}{d\beta} = -Y_1(1+\beta)^{-2} < 0$$ ie. the more the HH cares abt the future, the less it consumes today

$$\frac{dc_1^*}{dY_1} = \frac{1}{1+\beta} > 0$$ ie. the more income HH has, the more HH consumes today

$$\frac{dc_2^*}{d\beta} = \frac{1}{2}\left(\frac{\beta Y_1}{1+\beta}\right)^{-1/2} \left[\frac{Y_1(1+\beta) - \beta Y_1}{(1+\beta)^2}\right] > 0$$ ie. more HH cares abt the future, the more HH consumes tomorrow

$$\frac{dc_2^*}{dY_1} = \frac{1}{2}\left(\frac{\beta Y_1}{1+\beta}\right)^{-\frac{1}{2}} \frac{\beta}{1+\beta} > 0$$ ie. more income HH has, the more HH consumes tomorrow

ex / social security

$$\max_{S, c_1, c_2} u(c_1, c_2) \quad \text{s.t. } c_1 + S = Y_1 (1-\tau)$$

$$c_2 = Y_2 + (1+r) S + T$$

Govt BC: $Y_1 \tau = T$

- solve r^*, S^*, K^*, Y^{s*}, c_1^*, c_2^* in terms of Y_1, β, & τ

Chapter 7: Unemployment

February 24th.

* Unemployment

Def: - Frictional unemployment — unemployment resulting from workers being between jobs or entering or re-entering the labor force, ie "voluntary unemployment"

- Cyclical Unemployment — unemployment resulting from a recession or decrease in output

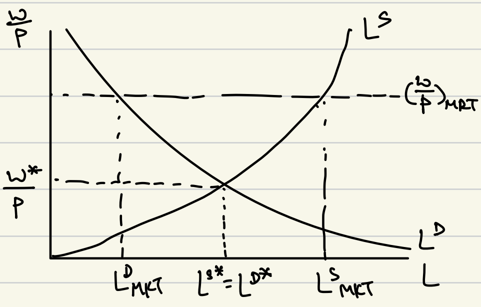
Def: structural unemployment

- (traditional) unemployment resulting when employees' skills do not match the skills desired by employers ex/ old industries dying out and new industries created, leaving previous workers with obsolete skills; AI, Textile workers in NC from international trade

(Farmer)

- unemployment caused by inefficient labor policies set by the government. ex/ - high minimum wages
  - high unemployment benefits
  - strict firing restrictions
  - powerful labor unions (public sector)

Labor Market



unemployment

$= L^S_{MKT} - L^D_{MKT}$

- What is unemployment in this graph?

$L^S > L^D \Rightarrow (\frac{W}{P})_{MKT} > \frac{W^*}{P}$

ie. one explanation of unemployment is a market wage above the equilibrium wage

- But why is $(\frac{W}{P})_{MKT} > \frac{W^*}{P}$?

one explanation is an "efficiency wage"

- Note: this cannot occur under perfect competition ; the firm is choosing its wage, not a price taker

* New Keynesian Theory of Unemployment

* Representative Firm - choose efficiency wage; given efficiency wage, max profits

Def: turnover costs - those costs incurred by hiring new workers to replace those workers who leave, ex/ - search costs, recruiting costs, training costs

$\Rightarrow$ cost of labor = real wage + turnover cost

- turnover cost Func: $C(\frac{W}{P}, L^D)$
- Assuming total turnover cost is proportional to the total number of workers

$$C(\tfrac{W}{P}, L^D) = \underbrace{c(\tfrac{W}{P})}_{\text{per worker turnover cost}} \underbrace{L^D}_{\text{\# of workers}}$$

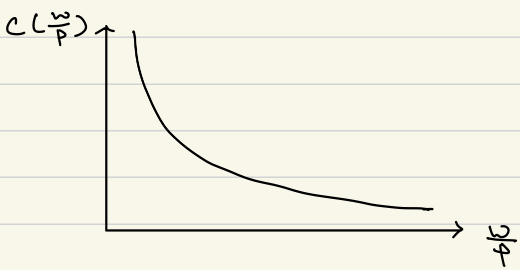
$$\frac{dC(\frac{W}{P}, L^D)}{dL^D} = c(\tfrac{W}{P}) > 0$$

$$\frac{dC(\frac{W}{P}, L^D)}{dc(\frac{W}{P})} = L^D > 0$$

- Assume $\frac{dc(\frac{W}{P})}{d\frac{W}{P}} = c'(\frac{W}{P}) < 0$

ie. As $\frac{W}{P}$ offered by the firm increases, workers are happier and less likely to quit job, and thus per worker turnover costs decreases

$c''(\frac{W}{P}) > 0$



Def: Efficiency Wage
- wage that minimizes total labor costs of the firm
ie. $(\frac{W}{P})_{eff}$

$\Rightarrow$ Firm wants to choose $\frac{W}{P}$ s.that.

Marginal Benefit per worker = Marginal Cost per worker

- solves:

$$\min_{\frac{w}{p}} \underbrace{\frac{w}{p}\cdot L^D + c\left(\frac{w}{p}\right)\cdot L^D}_{\mathcal{L}}$$

$$\frac{d\mathcal{L}}{d\frac{w}{p}} = L^D + L^D\cdot c'\left(\frac{w}{p}\right)$$

$$1 + c'\left(\frac{w}{p}\right) = 0$$

$$\Rightarrow \quad -c'\left(\frac{w}{p}\right) = 1$$ solve this to determine $\left(\frac{w}{p}\right)_{eff}$

ie. Marginal Benefit of $\frac{w}{p}$ per worker = Marginal Cost of $\frac{w}{p}$, per worker

- Given the efficiency wage, the representative firm maximizes profit:

$$\max_{Y^S, L^D} \quad Y^S - L^D\left(\frac{w}{p}\right)_{eff} - L^D c\left(\left(\frac{w}{p}\right)_{eff}\right) \quad \text{s.t. } Y^S = F(L^D)$$

$$\Rightarrow \max_{L^D} \quad \underbrace{F(L^D) - \left(\frac{w}{p}\right)_{eff} L^D - c\left(\left(\frac{w}{p}\right)_{eff}\right) L^D}_{\mathcal{L}}$$

$$\frac{d\mathcal{L}}{dL^D} = \underbrace{F'(L^D)}_{\substack{\text{Marginal product}\\ \text{or benefit of}\\ \text{labor}}} - \underbrace{\left(\frac{w}{p}\right)_{eff} - c'\left(\left(\frac{w}{p}\right)_{eff}\right)}_{\substack{\text{Marginal cost}\\ \text{of labor}}} = 0$$

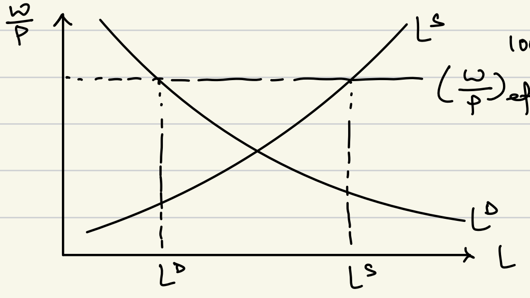
- Solve this to determine L^D

* Representative HH

- same as Chapter 4
- $L^S\left(\frac{w}{p}\right)$ given

* Natural Rate of Unemployment -

the amount of unemployment resulting from the efficiency wage, $\left(\frac{w}{p}\right)_{eff}$



$$100\% \cdot \frac{L^S - L^D}{L^S} = \text{natural rate of unemployment}$$

March 3rd

Practice:

ex/ turnover cost: $c(\frac{w}{p}) = \frac{.36}{\frac{w}{p}}$

production function: $Y^s = 1.6L^D - \frac{1}{2}(L^D)^2$

labor supply Func: $L^s = \frac{w}{p}$

Solve for $(\frac{w}{p})_{eff}$ and the natural rate of unemployment.

$$\min_{\frac{w}{p}} \frac{w}{p} \cdot L^D + c(\frac{w}{p}) \cdot L^D$$

$$= \frac{w}{p} \cdot L^D + \frac{.36}{\frac{w}{p}} \cdot L^D \qquad \frac{w}{p} = x$$

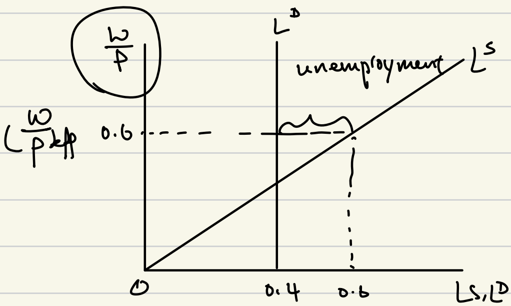
$$\frac{dL}{d\frac{w}{p}} = L^D - \frac{.36}{(\frac{w}{p})^2} \cdot L^D = 0$$

$$[L^D \cdot x + \frac{.36}{x} \cdot L^D]'$$

$$= L^D - \frac{.36}{x^2} \cdot L^D = 0$$

$$1 = \frac{.36}{x^2}$$

$$x^2 = 0.36 \quad \underline{x = 0.6} \quad (\frac{w}{p})_{eff} = 0.6$$



$$\text{Max } Y^s - L^D(\frac{w}{p})_{eff} - c((\frac{w}{p})_{eff}) \cdot L^D$$

$$= 1.6L^D - \frac{1}{2}(L^D)^2 - L^D \cdot 0.6 - \frac{0.36}{0.6} L^D$$

$$= 1.6L^D - \frac{1}{2}(L^D)^2 - L^D \cdot 0.6 - 0.6L^D$$

$$= (0.4L^D - \frac{1}{2}L^{D2})'$$

$$= 0.4 - L^D = 0$$

$$L^D = 0.4 \qquad L^S = 0.6$$

$$L^S - L^D = \underline{0.2}$$

Natural Rate of Unemployment $= \underline{0.2}/0.6$

* Chapter 8.

* New Keynesian AS

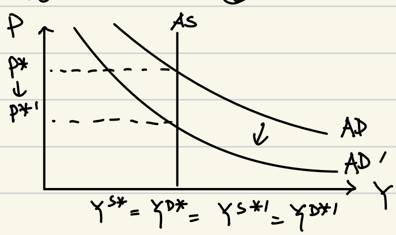
- Facts from the Great Depression
  1. unemployment (high) ~25% by 1933
  2. output, Y (low) ~ between 1929 to 1933 → 38% drop
  3. price level, P (low) ~ fall 24% between 1929 to 1933

- Trade as a share of GDP is small for US

"You're right, we did it. We're very sorry. But thanks to you, we won't do it again."
— Ben Bernanke, 2002
Friedman's 90th Birthday Celebration

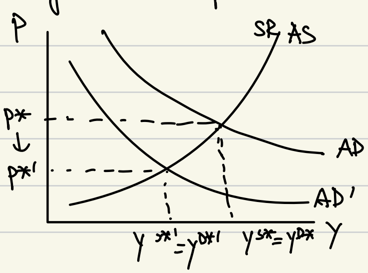
- Friedman and Schwartz, Monetary History of the United States

- Classical Model does not allow for ① (see ch7)
- What about ② and 3 ?



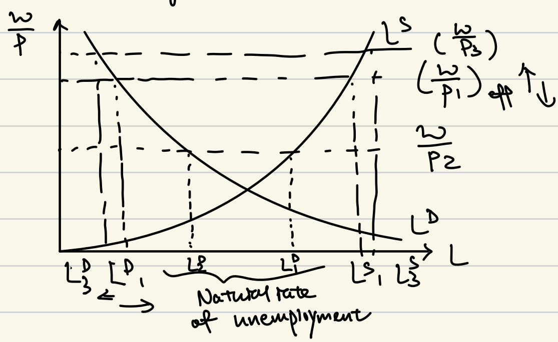
∴ Classical Model can match ③ but not ②.

- Keynesian Response to deliver lower Y and lower P?



- Keynes still believed ...

- But why is SRAS upward sloping?
  - Sticky wages: Firms set efficiency wage for workers (ch 7)
  - workers and firms set contracts for some time period at $(\frac{W}{P})_{eff}$
  - if P changes, firms cannot immediately adjust wages, W, because of contracts
  - changes in P change real market wage
- Note: Classical Model assumes nominal wages and prices adjust frictionlessly and instantaneously



Using sticky wages to derive SRAS:

- Note: L_1^D enters firm's production func to produce $Y_1^S = F(L_1^D)$

① Increase P_1 to P_2, that is $P_2 > P_1$

$\Rightarrow \frac{W}{P_2} < (\frac{W}{P_1})_{eff}$

$\Rightarrow L_2^D > L_1^D$

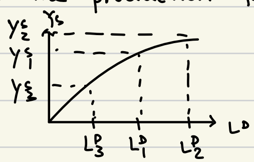
② Decrease P_1 to P_3, $P_3 < P_1$

$\Rightarrow \frac{W}{P_3} > (\frac{W}{P_1})_{eff}$

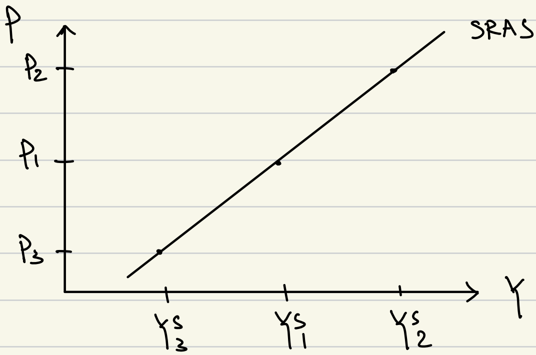
$\Rightarrow L_3^D < L_1^D$

$\therefore$ ① + ② $\Rightarrow L_3^D < L_1^D < L_2^D$ and $P_3 < P_1 < P_2$

Using the production func:



$\Rightarrow Y_3^S < Y_1^S < Y_2^S$



- Note: neutrality of money does not hold

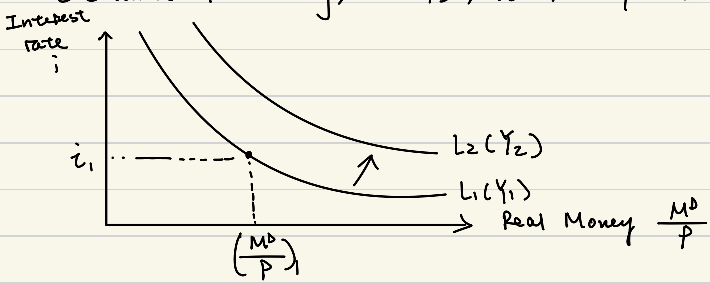
* IS - LM and Keynesian AD

(ch11) IS - LM (ch9)

- IS: Investment = Saving; Capital Market
- IS - LM → AD (ch12)
- LM: Money Market

* Chapter 9. Demand for Money and LM Curve

- Motivation: Keynesian AD/AS or "demand management" whereas Classical AD/AS only changes with changes in M^S; drop some classical assumptions to reformulate theory of AD
- Drop classical assumption of K (propensity to hold money) being constant; interest rate's influence on K
- cash vs. bonds ⇒ $\underbrace{\text{marginal utility of real money}\left(\frac{M^D}{P}\right)}_{\text{Benefit of money to conduct transactions}} = \underbrace{\text{nominal interest rate}}_{\text{cost of money b/c could've put in bonds}}$
- Demand for money, $L(Y)$, where Y = income

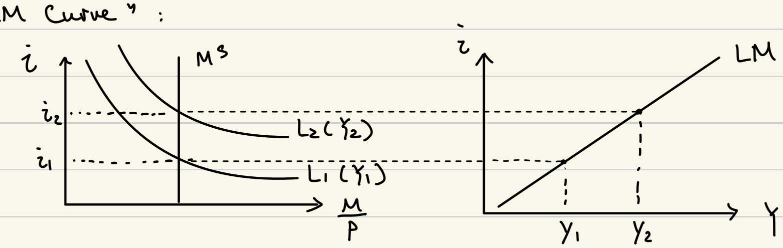


$Y_1 < Y_2 \Rightarrow L_1 < L_2$ b/c HH needs more cash for more transactions

Note: $M^D/P = \frac{h}{i} Y$ where h is now Keysian propensity to hold money, $K = h/i$
(as $i \uparrow$ $\frac{M^D}{P} \downarrow$)

* LM Curve

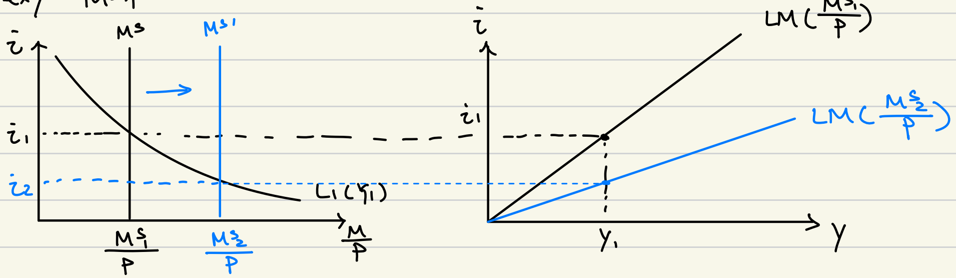
- Assume M^s fixed, Assume P fixed. Derive relation b/n interest rate and income, the "LM Curve":



- Note: at every point on LM, $M^s = M^D$, equilibrium in the money market
- Algebra: combining $M^s = M^D$ and $\frac{M^D}{P} = \frac{k}{i} Y \Rightarrow i = \frac{kP}{M^s} \cdot Y$ "LM Curve"

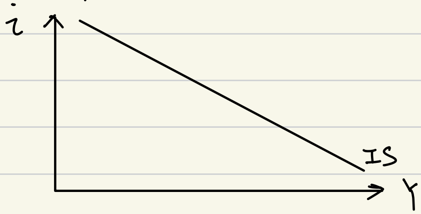
* Shifters of the LM Curve

ex/ $M^s \uparrow$



* Chapter 11 -- IS Curve

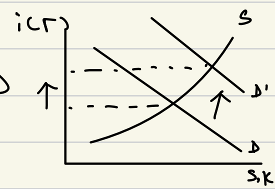
- Recall two-period model's "level effect": Y_1 increases → Savings increases → r^* decreases
  ie. Higher income drives down the interest rate, which is the "IS Curve"



- Note: at every point on IS, S = K ie savings = investment, equilibrium in the capital market

- Notice a problem? $i \neq r$ ie. nominal ≠ real interest rate
  $r = i -$ inflation

- Shifts in the IS Curve (for a fixed level of income, Y)

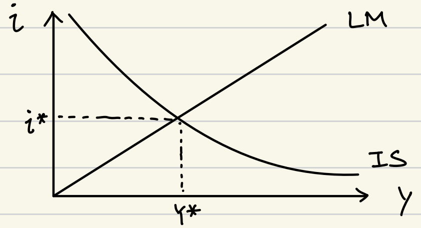


1) Increase in Government purchases (assume no effect on HH savings)
   ⇒ i increases b/c government purchases = investment demand
   ⇒ IS ↑
   Note: ignores "crowding out"
2) Increase Taxes
   1) Government needs less finance from capital market so gov investment decreases and then i decreases
   2) HHs poorer so savings decreases, and then i increases
   ∴ move in IS curve depends on the sizes of 1) and 2)
   - Keynesian typically argue 1) > 2) ⇒ IS decreases

Note: "Ricardian Equivalence" ⇒ 1) = 2) so IS doesn't change
(HH realizes it's poor & make decisions to offset government decisions)

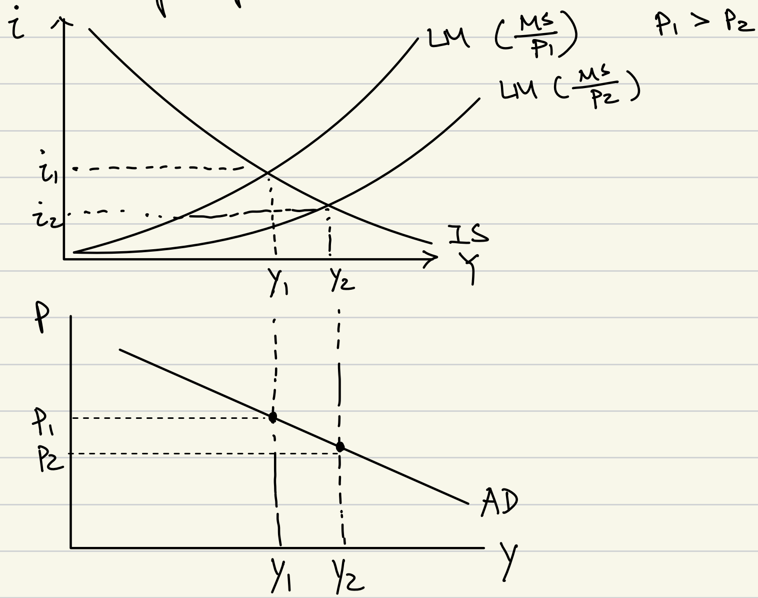
3) Technological change / productivity increase
   ⇒ investment increases then i increases
   ⇒ IS increases

* Chapter 12: IS-LM and Keynesian Aggregate Demand



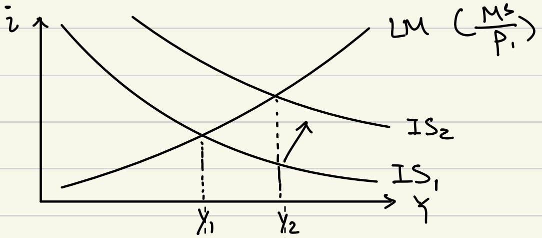
Note: point of intersection is IS-LM equilibrium where money market and the capital market are in equilibrium

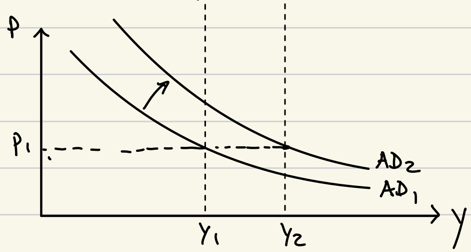
- Deriving Keynesian AD → Δ price in IS-LM, see how income Y changes



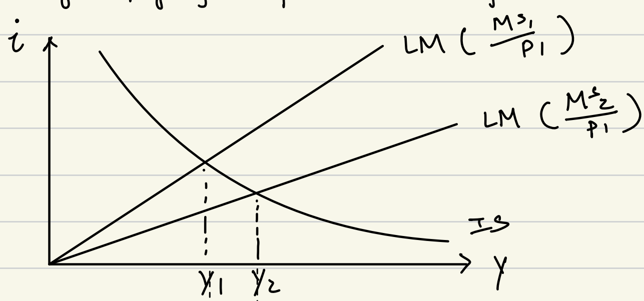
- Shifts in AD

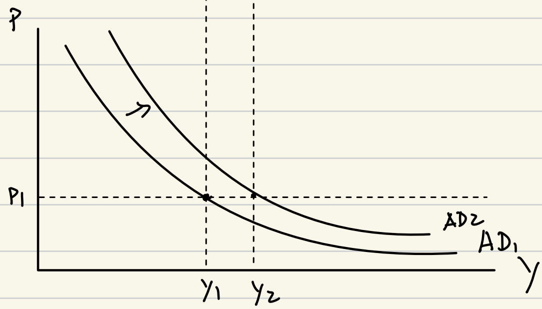
1) Gov't purchases go up then IS goes up then AD goes up



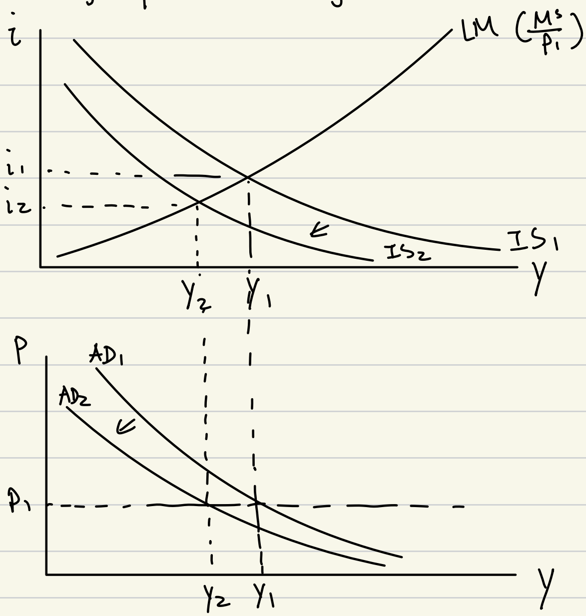


2) Money supply goes up then LM goes down then AD goes up





3. Taxes go up then IS goes down then AD goes down



March 19th

* Supply of Money

Def: M1 is cash in the hands of the public, checking deposits, this is money in this course unless otherwise noted, M^s.

Def: M_2 is M_1 plus savings deposits.

Def: M_3 is M_2 plus everything else.

- Money in the macroeconomy -

3 agents - ① treasury

② Central Bank (Federal Reserve in the US, European Central Bank, Bundesbank, Bank of England, Hong Kong Monetory Authority etc.)

③ Commercial Banks (Bank of America, Truist etc.)

1. Treasury
   - collects tax revenue & disburses money to finance government expenditure
   - when tax revenue < gov't expenditure, need to issue government debt
   - sell govt's bonds in the capital market, ie. borrow money to be paid back w/interest in the future (in US, "T-bills")

2. Central Bank

   Richmond, New York, St. Louis, Atlanta, Philadelphia, Chicago, Boston, San Francisco, Minneapolis, Dallas, Kansas City, Cleveland
   - nation's principle monetary authority
   - affects supply of money through 3 traditional mechanisms:
     1) Required reserve ratio (RRR)
        - minimum fraction of deposits that commercial banks must hold in the form of cash
        - ex/ RRR = 10%
          - Someone deposits \$100, banks must keep \$10
          - RRR ↑ ⇒ M^s ↓   RRR ↓ ⇒ M^s ↑

2) Lender of the last resort

- lends money to commercial banks at the "discount rate" if banks need resources
  - Moral Hazard

3) Open Market Operations

- primary mechanism
- purchases gov't bonds in capital, ie. money flows into the capital market and $M^s \uparrow$
- Sells govt bonds in capital market ie. takes money out of capital market, and $M^s \downarrow$

① The supply of money (Ch10)

② Difference Equations

3. Commercial Banks

- hold money (deposits)
- lend money (loans)
- reserves - money not lent out
- required reserves = RRR · deposits ie. minimum amount of reserves banks must hold
- creating M1

ex1/ Bank A

RRR = 10%

person 1 deposits \$100 ⇒ M1 = \$100

person 2 gets a \$90 loan ⇒ M1 = \$190 (100+90)

Bank B

person 2 spends money in person 3's store & person 3 deposits \$90

\$90 ⇒ M1 = 190\$ (100+90)

Person 4 gets a \$81 loan : M1 = 100+90+81 = \$271

- this process continues...

M1 = 100

but through commercial bank lending...

$$M1 = 100 + 100(1-RRR) + 100(RRR)^2 + \cdots$$

$$= \frac{100}{1-(1-RRR)} = \frac{100}{1-0.9} = 1000$$

by property of geom theories

- excess reserves
- secular stagnation – economy run out in tech improvements, slow down in growth

* Money Multiplier, Monetary Base, and Money Supply

Def. Monetary Base (MB) – the amount of money central bank has put into the market through the purchase of government bonds; central bank controls this amt entirely

Def: Money Supply (M^S) is the actual amount of M1; central bank and commercial banks control this amount

$$MB = \underset{\substack{\uparrow \\ \text{reserves held} \\ \text{by the commercial;} \\ \text{"controlled" by setting} \\ \text{RRR}}}{R} + CU \rightarrow \text{cash or currency held by the public}$$

$$M^S = \underset{\substack{\uparrow \\ \text{Deposits}}}{D} + CU$$

* $R < D \Rightarrow MB < M^S$

- claim: MB and MS are proportional, i.e. there exists some m s.t. $mMB = M^S$, where m is the "money supply multiplier"
- Deriving m

$$MB = R + CU$$

$$\frac{MB}{D} = \frac{R}{D} + \frac{CU}{D}$$

define $rd = \frac{R}{D} \rightarrow$ ratio of reserves to deposits

$cu = \frac{CU}{D} \rightarrow$ ratio of currency to deposits

$$\frac{MB}{D} = rd + cu$$

$$M^S = D + CU$$

$$\frac{M^S}{D} = \frac{D}{D} + cu$$

$$\frac{M^S}{D} = 1 + cu$$

$$\frac{M^S/D}{MB/D} = \frac{1+cu}{rd+cu} = \frac{M^S}{MB} = \underline{m}$$

$$M^S = \underbrace{\left(\frac{1+cu}{rd+cu}\right)}_{=m} MB$$

$\Rightarrow M^S = m \cdot MB$

$\therefore$ Given knowledge of rd and cu, central bank can also indirectly control M^S

* Chapter 14: Difference Equations

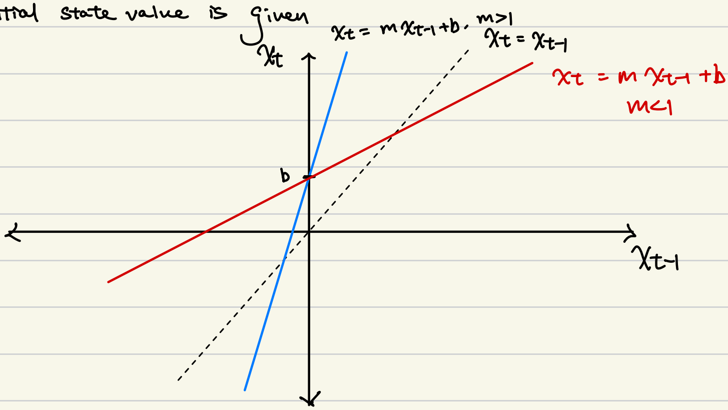
Def: difference equation — an equation showing how one variable (called a state variable) changes from period to period depending on the variable's own past values and some parameters

ex/ difference equation:

① $x_t = m x_{t-1} + b$ where b is the y-intercept and m is the slope and x is the state variable

② 45° line : $x_t = x_{t-1}$ steady state condition, where $b=0$ and $m=1$

③ x_0 ie. initial state value is given



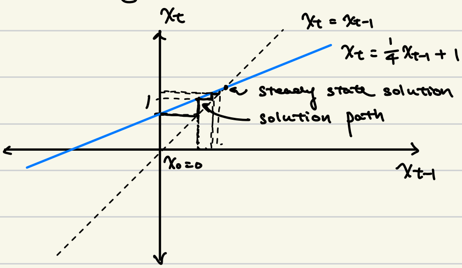
Def: Solution to difference equations — a list of values for the state variable: one for each period t, for a given initial value x_0

ex./ $x_t = \frac{1}{4} x_{t-1} + 1$

let $x_0 = 0$ then

x	solution
x_0	0
x_1	1
$x_2 = \frac{1}{4}(1)+1 = 1.25$	
$x_3 = \frac{1}{4}(1.25)+1 = 1.31$	
$x_4 = \frac{1}{4}(1.31)+1 = 1.33$	
$x_5 = \frac{1}{4}(1.33)+1 = 1.33$	

Graphically ...



Def: solution path of the difference equation – the graphical representation tracing the solution to the difference equation

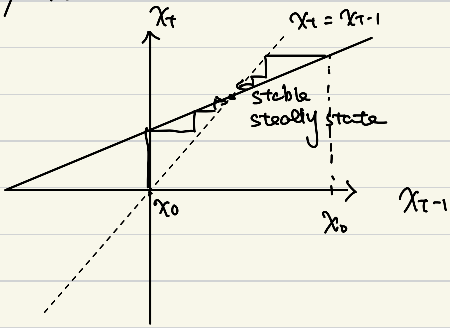
Def: steady state solution – the intersection of difference equation and the 45°, or steady state condition line

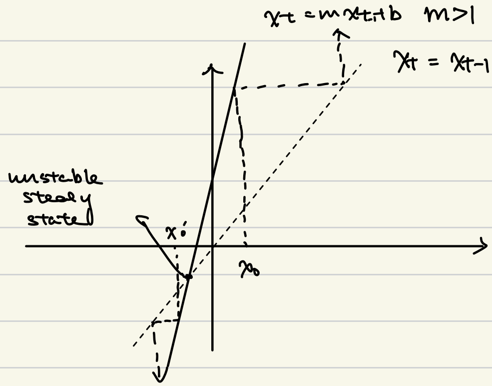
ex./ $x_t = m x_{t-1} + b$, $x_t = x_{t-1} = x_{ss}$

$$x_{ss} = m \cdot x_{ss} + b$$
$$x_{ss} - m x_{ss} = b$$
$$x_{ss}(1-m) = b$$
$$x_{ss} = \frac{b}{1-m}$$

Def: stable steady state – when the solution to the difference equation converges to a steady state, that steady state is called "stable"

ex/ $x_t = m x_{t-1} + b$, $m < 1$





Def: unstable steady state – when the solution to the difference equation diverges away from the steady state, that steady state is called an "unstable" steady state

March 26th

① Dynamics of Government debt

② Midterm 2 Next Thursday

* Dynamics of Govt Debt

$$B_t = (1+r)\, B_{t-1} + D_t V_t$$ where r is the nominal interest rate, $r_t = rV_t$ by assumption

D_t: Government Deficit

B_t is the current nominal government debt

B_{t-1} is the previous nominal government debt

D_t is the primary govt budget deficit ie. govt outlays − govt revenue

- Who cares what the level is?
- Need a better measure;

$\frac{\text{debt}}{\text{GDP}}$ b/c gov't can tax GDP to repay debt

- Reinhart & Rogoff (2009) Finds real GDP growth not affected w/ $\frac{\text{debt}}{\text{GDP}}$ below 90%
- Debt to GDP Ratio

$$\frac{B_t}{P_t Y_t} = (1+r)\frac{B_{t-1}}{P_t Y_t} + \frac{D_t}{P_t Y_t} V_t$$ by dividing by nominal GDP

RHS does not conform with the difference equation.

$$\frac{B_t}{P_t Y_t} = (1+r)\frac{B_{t-1}}{P_{t-1} Y_{t-1}} \cdot \frac{P_{t-1} Y_{t-1}}{P_t Y_t} + \frac{D_t}{P_t Y_t}$$

Recall: $g_t = \frac{P_t Y_t - P_{t-1} Y_{t-1}}{P_{t-1} Y_{t-1}}$ $\quad 1+g_t = \frac{P_t Y_t}{P_{t-1} Y_{t-1}} \rightarrow \frac{P_{t-1} Y_{t-1}}{P_t Y_t} = \frac{1}{1+g_t}$

$$\Rightarrow \frac{B_t}{P_t Y_t} = (1+r)\frac{B_{t-1}}{P_{t-1} Y_{t-1}} \cdot \frac{1}{1+g_t} + \frac{D_t}{P_t Y_t}$$

$$\Rightarrow b_t = \frac{(1+r)}{(1+g)} \cdot b_{t-1} + d$$

where $g_t = g \; \forall t$ by assumption,
$d_t = d \; \forall t$ by assumption

$$b_t = \frac{(1+r)}{(1+g)} b_{t-1} + d$$

$$\parallel \quad \parallel \quad \parallel \quad \parallel$$

$$x_t = m \cdot x_{t-1} + b$$

- Note:
  we treat $r, g,$ & d as indepenedent but not necesscrily the case in the real world
- Note:
  importance of interaction b/n r and g
  when $d = 0$, $b_{t-1} > 0$
  if $r > g$ then $b_t \uparrow$
  if $r < g$ then $b_t \downarrow$

in general, r puts upward pressure on b_t
g puts downward pressure on b_t

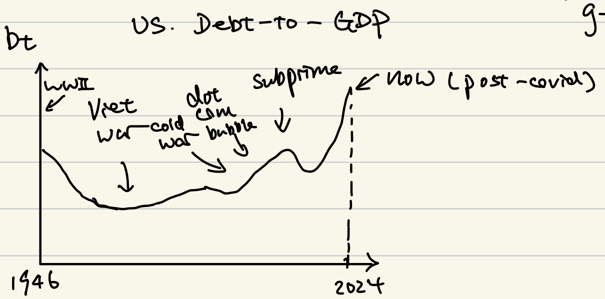
- Note: $d \uparrow$ then $b_t \uparrow$ ie. $\frac{\text{deficit}}{\text{GDP}} \uparrow \frac{\text{debt}}{\text{GDP}}$
- steady state b_{ss}

$$\left.\begin{aligned} b_t &= \frac{1+r}{1+g} b_{t-1} + d \\ b_t &= b_{t-1} = b_{ss} \end{aligned}\right\} \Rightarrow \quad b_{ss} = \frac{1+r}{1+g} b_{ss} + d$$

$$\vdots$$

$$b_{ss} = \frac{1+g}{g-r} d$$

US. Debt-to-GDP

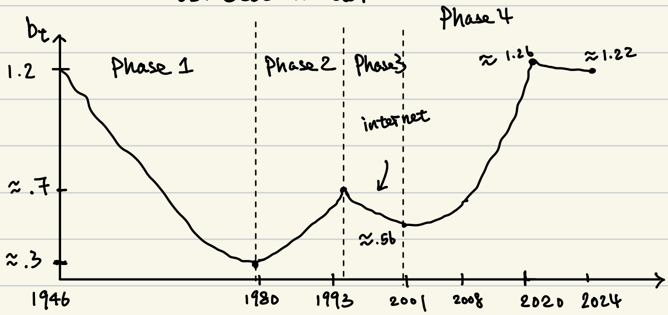


March 31st

- Dynamics of Govt Debt

g = growth rate

US. Debt to GDP



	Phase 1	Phase 2	Phase 3
Avg d	−1.2%	.7%	−1.9%
Avg g	7.5%	6.8%	5.5%
Avg r	4.1%	7.5%	4.9%
$\frac{(1+r)}{(1+g)}$	.97	1.01	.99

$b_t = .97\, b_{t-1} - .012$ $\quad b_t = 1.01\, b_{t-1} + .007$ $\quad b_t = .99\, b_{t-1} - 0.19$

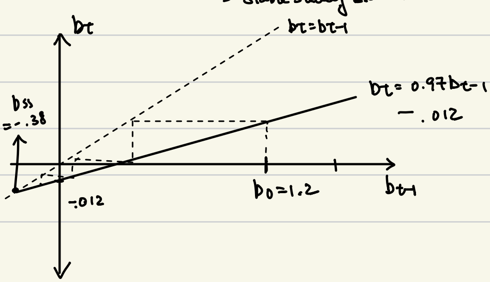
Phase 1	Phase 2	Phase 3
$b_{ss} = \frac{1+.075}{.075-.041} \cdot (-.012) \approx -.38$	$b_{ss} \approx -1.07$	$b_{ss} \approx -3.34$

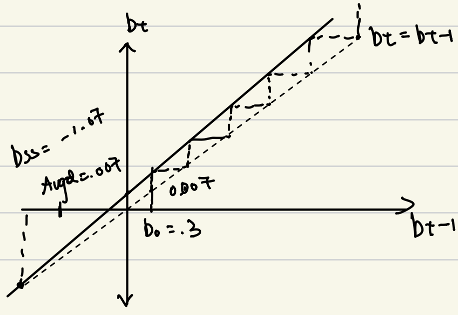
→ unstable steady state

Phase 2 $b_t = 1.01\, b_{t-1} + .007$

Phase 1 = $b_t = .97 b_{t-1} - .012$

↳ stable steady state

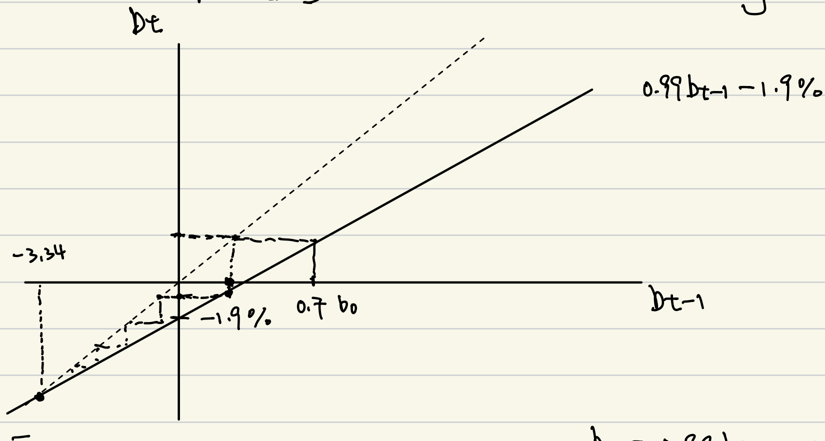




Phase 3

$d = -1.9\%$ $g = 5.5\%$ $r = 4.9\%$

$\frac{(1+r)}{(1+g)} = 0.99$



$b_t = 0.99\, b_{t-1} - 0.019$

$0.01\, b_t = -0.019$

$b_t = -1.9$

$$b_{ss} = \frac{1.04}{1.06} b_{ss} + 0.02$$

g: 1.9%

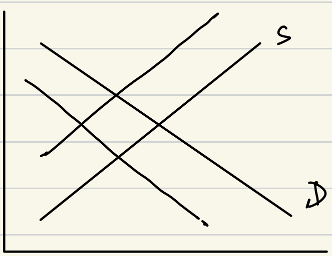
$$b_{ss} = \frac{1+r}{1+g} b_{ss} + d$$

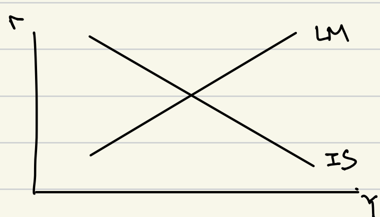
deficit = 3.20%

$$b_{ss} = \frac{1+g}{g-r} d$$

$$= \frac{1+6\%}{6\% - 4\%} d$$

Tax Cut → G↓

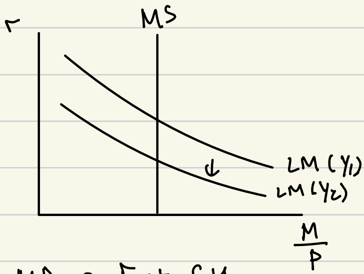


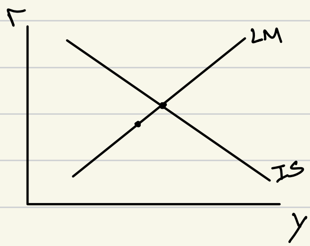


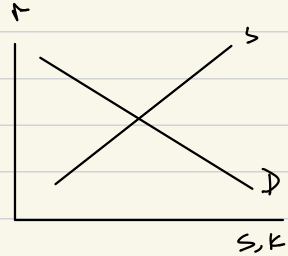
if G increase → higher demand for investment

higher r → higher savings

$b_t = m b_{t-1} + d$



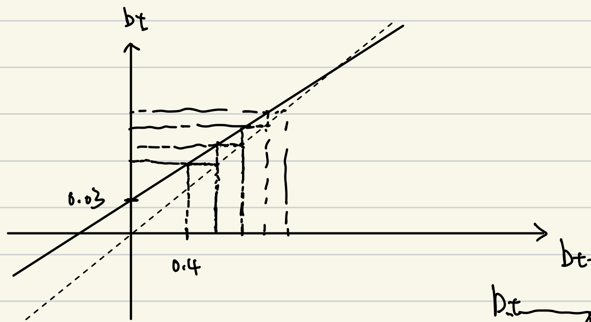




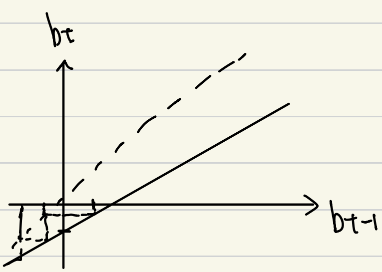
$MB = r + cu$

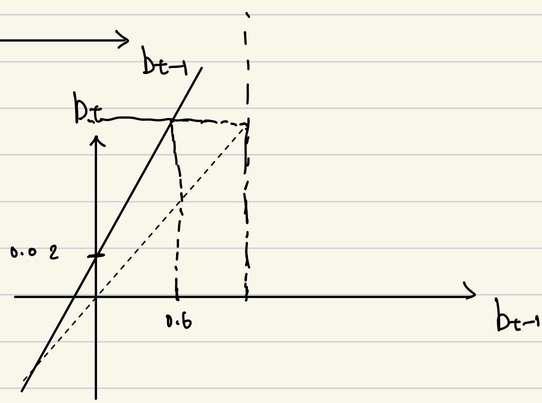
$MS = D + cu$

$\frac{MB}{D} = rd + cu$ $\qquad$ $\frac{MS}{D} = 1 + cu$



$$b_t = \frac{1+0.02}{1+0.05} b_{t-1} + 0.03$$





$b_{ss} = 1.5 b_{ss} + 1$ $\qquad$ $b_{ss} = 1.5 b_{ss} - 1$

$-0.5 b_{ss} = 1$ $\qquad$ $-0.5 b_{ss} = -1$

$b_{ss} = -2$ $\qquad$ $b_{ss} = 2$

Q1.

$c(\frac{w}{p}) = -\ln(\frac{w}{p}) \qquad Y^s = 2(L^D)^{\frac{1}{2}}$

a) Efficiency wage: $\frac{d}{d\frac{w}{p}}(\frac{w}{p} \cdot L^D - \ln(\frac{w}{p})L^D)$

$= \cancel{L}^D = \frac{\cancel{L}^D}{w/p} \qquad \frac{w}{p} = 1$

b) $L^S = 2(\frac{w}{p})$

$(2(L^D)^{\frac{1}{2}} - \frac{w}{p}L^D + \ln(\frac{w}{p})L^D)'$

$L^S = 2$

$= L^{D-\frac{1}{2}} - \frac{w}{p} + \ln(\frac{w}{p}) = 0$

$L^{D-\frac{1}{2}} = 1$

NRU $= 2-1=1$

$L^D = 1$

2. $-\frac{d}{d\frac{w}{p}}(4(\frac{w}{p})^{-1}) = 4(\frac{w}{p})^{-2}$

$(\frac{w}{p}L^D + 4(\frac{w}{p})^{-1}L^D)'$

$= L^D = 4(\frac{w}{p})^{-2}L^D$

$2(\frac{w}{p})^{-1} = \frac{w}{p}$

$2 = (\frac{w}{p})^2$

$(\frac{w}{p})^{-2} = \frac{1}{4}$

$\frac{w}{p_{eff}} = 2$

$\frac{w}{p_{MKT}} = \sqrt{2}$

$(\frac{w}{p})^{-1} = \frac{1}{2}$

$MB = r + cu$

$M^s = d + cu$

$rd = \frac{r}{D}$

$cu = \frac{cu}{d}$

$\frac{M^s/D}{MB/D} = \frac{1+cu}{rd+cu} = m$

$M^s = m \cdot MB$

$b_t = \frac{1+r}{1+g} b_{t-1} + d$

$b_{ss} = \frac{1+g}{g-r} d$

April 7th.

① Knowledge and the Wealth of Nations by David Walsh

② Economic Growth: Motivation & Data (ch 15)

* Economic Growth

- Why are some countries rich and others poor?

  technology "barriers", natural resources, war / conflict, institutions - democracy (?) / Form of govt, education / human capital, Geography, rule of Law, contract enforcement

  health, culture, religion

  economic institutions — market vs govt intervention, central bank

  Banking / Financial Sector → religion: X charge high interest rates

  Tourism / trade, exchange rate policy

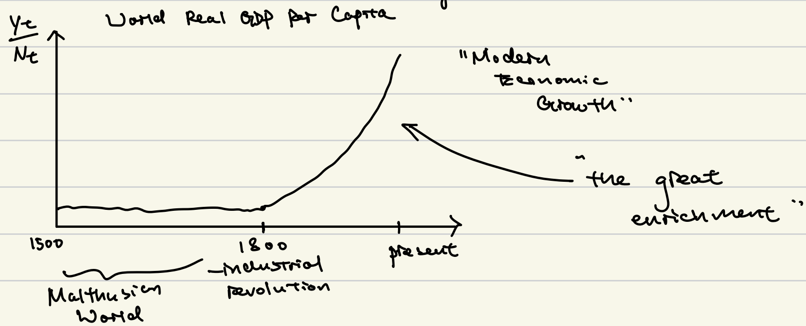
  people, physical capital

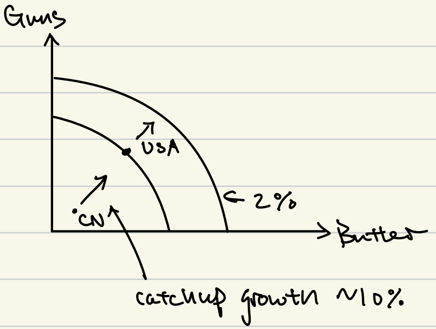
  Foreign aid ("Dead Aid")

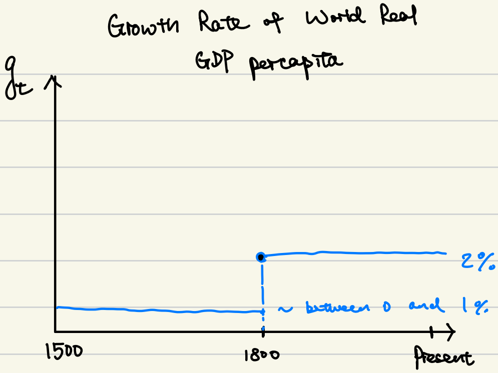
  infrastructure

{ infrastructure

→ "TFP" → Total Factor Productivity







- higher Real per Capita GDP ~ lower growth rate
- contagion 1997
- 1990s → Japan → Housing Bubble
- Liquidity Trap
- India — strict labor laws

- Botswana → ~3800% Growth

April 14th.

① Economic Growth (ch 15)

② The Bottom Billion By Paul Collier

* Economic Growth

- exogenous growth theory (ch 15) — growth theory that does not explain long-run economic growth by forces within a model, that is, long-run growth is assumed to occur exogenously
- endogenous growth theory (ch 16) — growth theory explaining long-run economic growth by forces within a model

* Cobb-Douglas Production Function

$$Y = AK^a(QL)^{1-a}$$ only inputs

where L = labor, K = Capital, a measures relative importance of K or L, $a \in [0,1]$

Q is efficiency of labor, A is a constant

- Constant Return to Scale:

$f(\lambda K, \lambda L) = \lambda f(K, L) \quad \forall \lambda > 0$

Claim: Cobb-Douglas is CRS

Proof. $A(\lambda K)^a (\lambda Q L)^{1-a}$

$= AK^a (QL)^{1-a} \lambda^a \lambda^{1-a} = \lambda A K^a (QL)^{1-a}$

- Recall: $\dfrac{df(K,L)}{dK} \equiv MPK$

$$\frac{df(K,L)}{dL} \equiv MPL$$

$\therefore$ $MPK = aAK^{a-1}(QL)^{1-a}$

$MPL = (1-a)AK^{a}Q^{1-a}L^{-a}$

- Showing relation b/n a & $(1-a)$ in model w/ data:

Representative Firm's problem:

$$\max_{Y,K,L} \; P\cdot Y - w\cdot L - r\cdot K \quad \text{s.t. } Y = AK^{a}(QL)^{1-a}$$

$$\Rightarrow \max_{K,L} \underbrace{P\cdot AK^{a}(QL)^{1-a} - w\cdot L - r\cdot K}_{\mathcal{L}}$$

$$\frac{d\mathcal{L}}{dK} = apAK^{a-1}(QL)^{1-a} - r = 0$$

$$\Rightarrow ① \; \frac{r}{P} = aAK^{a-1}(QL)^{1-a} = MPK$$

$$\frac{d\mathcal{L}}{dL} = (1-a)pAK^{a}Q^{1-a}L^{-a} - w = 0$$

$$\Rightarrow ② \; \frac{w}{P} = (1-a)AK^{a}Q^{1-a}L^{-a} = MPL$$

- Divide ① by Y:

$$\frac{r}{PY} = \frac{aAK^{a-1}(QL)^{1-a}}{Y}$$

$$\frac{r}{PY} = \frac{aAK^{a-1}(QL)^{1-a}}{AK^{a}(QL)^{1-a}} = aK^{-1}$$

$$\underbrace{\frac{rK}{PY}}_{\text{Capital's Share of Income, GDP}} = a$$

- divide ② by Y:

$$\frac{w}{PY} = \frac{(1-a)AK^{a}Q^{1-a}L^{-a}}{AK^{a}(QL)^{1-a}}$$

$$\frac{w}{PY} = \frac{(1-a)}{L} \qquad \underbrace{\frac{wL}{PY}}_{\text{Labor share of income, GDP}} = 1-a$$

- Data $\Rightarrow a, 1-a$

$(1-a) \in [.6, .8]$

* Growth Accounting

Def: Growth Accounting – Method of Calculating the shares of growth in real GDP per Capita that are attributable to growth in capital, labor, and Total Factor Productivity (TFP)

- Mapping Cobb-Douglas to the data:

CD: $Y = A K^a (QL)^{1-a}$

Data: $\frac{Y}{N}$ ie. real GDP per capita, where N = population

CD: $\frac{Y}{N} = \frac{AK^a (QL)^{1-a}}{N}$

$\frac{Y}{N} = (AQ^{1-a}) \frac{K^a L^{1-a}}{N^a N^{1-a}}$ where $AQ^{1-a} \equiv$ TFP or Solow Residuals

$$\underbrace{\frac{Y}{N}}_{\text{data}} = (AQ^{1-a}) \underbrace{\left(\frac{K}{N}\right)^a}_{\text{data}} \underbrace{\left(\frac{L}{N}\right)^{1-a}}_{\text{data}}$$

$$\frac{\frac{Y}{N}}{\left(\frac{K}{N}\right)^a \left(\frac{L}{N}\right)^{1-a}} = AQ^{1-a}$$

- You need increase in productivity to see increase in income

① Economic Growth (ch15); practice problem set online (not #4)

* Neoclassical Growth Model

$X \rightarrow$ investment

③ $S_t = X_t \; \forall t$ ie. Savings = investment, by def recall NIPA Model 2

④ $Y_t = A K_t^a (Q_t L_t)^{1-a} \forall t$ ie. Cobb-Douglas

⑤ $K_{t+1} = (1-\delta) K_t + X_t \; \forall t$ ie. Law of Motion of Capital

⑥ $\frac{S_t}{Y_t} = s \forall t$ ie. savings rate is constant

Assume:

A1: L is inelastically supplied $\Rightarrow \frac{L}{N} = 1$ by $L = N$

A2: population is constant, $N_{t+1} = N_t \ \forall t$

A3: efficiency of labor is constant, $Q_t = 1 \ \forall t$

$\Rightarrow Y_t = AK_t^{a} L_t^{1-a} \ \forall t$

- Recall, we want everything in per capita terms:

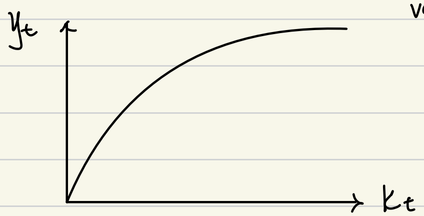
$$\frac{Y_t}{N_t} = \frac{AK_t^a L_t^{1-a}}{N_t}$$

$$\frac{Y_t}{N_t} = \frac{AK_t^a L_t^{1-a}}{N_t^a N_t^{1-a}}$$

$$\frac{Y_t}{N_t} = A\left(\frac{K_t}{N_t}\right)^a \left(\frac{L_t}{N_t}\right)^{1-a} \qquad a=1$$

$$\frac{Y_t}{N_t} = A\left(\frac{K_t}{N_t}\right)^a$$

$\Rightarrow$ ⑦ $y_t = Ak_t^a$ ie. "per capita production function where lower case variables represent per capita terms"



$$\frac{dy_t}{dk_t} = aAk_t^{a-1}$$

$$\frac{d^2y_t}{dk_t^2} = (a-1)aAk_t^{a-2}$$

Diminishing Marginal product of capital as a property of the per capita production function

- Note: ⑦ shows y_t for a given k_t but does not show how y_t evolves, but we know how k_t evolves

- Note: Dividing ③-⑥ by N_t gives ③'-⑥' in per capita terms

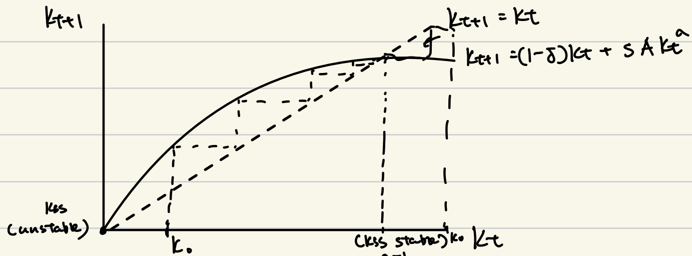
$\Rightarrow$ ⑤' $k_{t+1} = (1-\delta)k_t + x_t$

$k_{t+1} = (1-\delta)k_t + s_t$ by ③'

$k_{t+1} = (1-\delta)k_t + sy_t$ by ⑥'

✱ ⑧ $k_{t+1} = (1-\delta)k_t + sAk_t^a$ by ⑦

"per capita neoclassical growth equation"



$$\frac{dk_{t+1}}{dk_t} = (1-\delta) + asAk_t^{a-1} > 0$$

$$\frac{d^2k_{t+1}}{dk_t^2} = (a-1)\alpha sAk_t^{a-2} < 0$$

- if $k_0 < k_{ss}$ then economy produces $y_0 = Ak_0^a$ and invests $s \cdot Ak_0^a$ to add to capital stock to get $k_1 > k_0$, which implies $y_1 > y_0$; however, diminishing marginal product of capital sets in, steady state is reached, and growth stops
- if $k_0 > k_{ss}$ then because of diminishing marginal product of capital, new investment is not large enough to offset depreciation of capital stock, $k_1 < k_0$ so $y_1 < y_0$; steady state reached, and growth decline stops

$\Rightarrow g_{k_t} \rightarrow 0$ ie. growth rate of capital per capita

$g_{y_t} \rightarrow 0$ ie. growth rate of output/income per capita

① Economic Growth (Ch 15); practice problem set online

② Endogenous Growth Theory (Ch 16)

- Steady State K_{ss}:

$$⑧ \Rightarrow k_{ss} = (1-\delta) k_{ss} + sAk_{ss}^{a}$$

$+ \; k_{t+1} = k_t = k_{ss}$

$\vdots$

$$k_{ss} = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-a}}$$

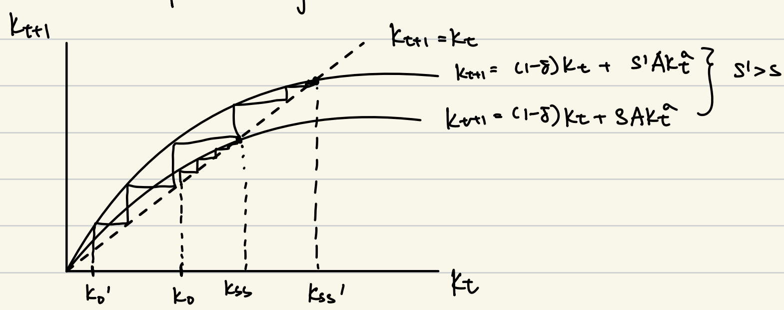
- Note: now we know how to exogenously increase or decrease k_{ss} in order to increase or decrease y_{ss} where $y_{ss} = Ak_{ss}^{a}$

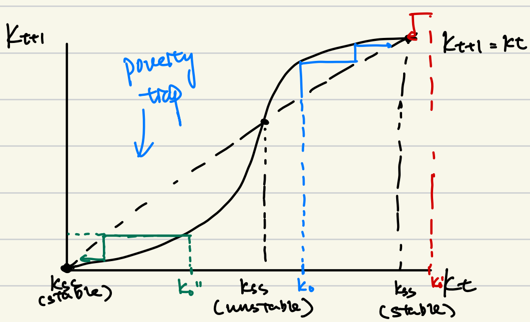
- change in savings rate, s

$$\frac{dk_{ss}}{ds} = \frac{1}{1-a}\left(\frac{A}{\delta}\right)^{\frac{1}{1-a}} \cdot s^{\frac{a}{1-a}} > 0$$

ie. A higher savings rate increases k_{ss}

• Changes in A





* Endogenous Growth Theory

exogenous growth theory

$\Rightarrow A\ (Q_t)$ grows exogenously

endogenous growth theory

$\Rightarrow$ Human Capital / Knowledge accumulation as an explanation for growth in Q_t

- knowledge / human capital accumulation

① investment (like physical capital) like education

② Learning by doing - knowledge acquired through the act of production

- knowledge as an externality

Def: positive externality - benefit gain by an economic agent without that agent paying for it

① $S_t = X_t \ \forall t$ ie. savings = investment

② $Y_t = AK_t^a (Q_t L_t)^{1-a} \ \forall t$ ie. Cobb-Douglas

③ $K_{t+1} = (1-\delta) K_t + X_t \ \forall t$ ie. Law of Motion of Capital

④ $\frac{S_t}{Y_t} = s \ \forall t$ ie. Savings investment is constant

⑤ $Q_t = \frac{K_t}{N_t} \ \forall t$ ie. Learning by doing (productivity of each worker is proportional to capital per worker, which is a measure of industrialization)

Assume:

A1: L is inelastically supplied

$\Rightarrow \frac{L}{N} = 1$ by $L = N$

A2: population is constant, $N_{t+1} = N_t \ \forall t$

- Note: From ②, K_t^a is the "private effect" of capital accumulation, ie. benefit to the firm for accumulating capital on its own

- Taking into account of externality, using ⑤ and ②:

$$Y_t = AK_t^a \left(\frac{K_t}{N_t} L_t\right)^{1-a}$$

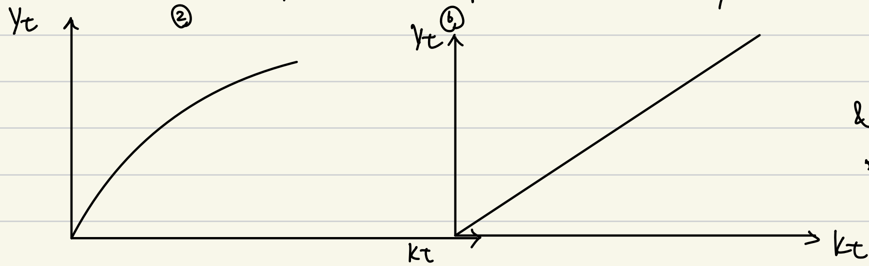
$$Y_t = AK_t^a K_t^{1-a} \text{ by A1}$$

Note: K_t^{1-a} is the "social effect" of capital accumulation, ie. benefit to the whole economy from the education of the work force by learning by doing

$\Rightarrow$ ⑥ $Y_t = AK_t$

Note: diminishing marginal product at the firm level in ②

constant marginal product of capital at the economy-wide level in ⑥



* This is the essential economic & mathematic mechanism in endogenous growth theory *

- Note: as in exogenous growth theory, we can make everything per capita ①' – ⑥'

$\Rightarrow$ ③' $k_{t+1} = (1-\delta)k_t + x_t$

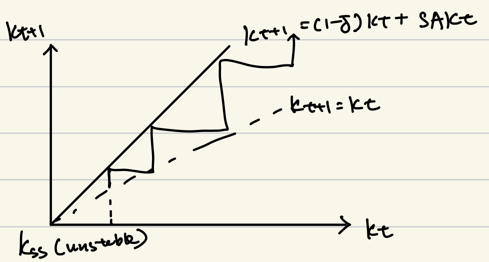
$k_{t+1} = (1-\delta)k_t + s_t$ by ①'

$k_{t+1} = (1-\delta)k_t + sy_t$ by ④'

* ⑦ $k_{t+1} = (1-\delta)k_t + sAk_t$ by ⑥'

where ⑦ is the "per capita endogenous growth equation"

$k_{t+1} = (1-\delta + sA)k_t$



- Note: Economy does not converge to a steady state and thus growth does not stop

- growth

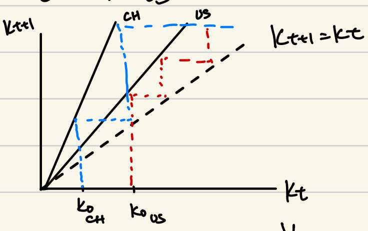
$$\frac{k_{t+1}}{k_t} = (1-\delta + sA) \text{ by } 7$$

$$1 + g_{kt} = 1 - \delta + sA$$

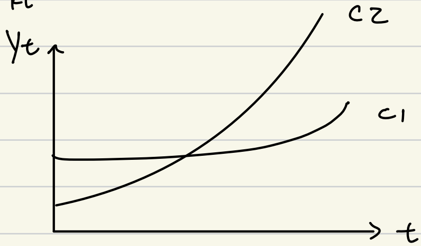
$$g_{kt} = sA - \delta$$

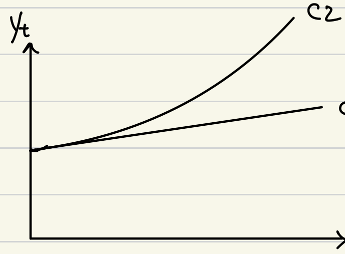
$$\frac{y_{t+1}}{y_t} = \frac{Ak_{t+1}}{Ak_t} \qquad 1 + g_{yt} = 1 + g_{kt} \qquad g_{yt} = g_{kt}$$

- China vs. US



- Change in A
- Change in δ





Final: Bring Calculators

Richer 1960 → countries X responsible for economic events

Growth Accounting
* Growth Disasters

· Europe vs. United States Capital · Labor, TFP

- How Govt Control Debt to GDP Ratio?
  Direct Control: Deficit → taxation & Spending

* Liquidity Trap – Growth Numericals
Japan (X need to know)